Complex Variable

Example 03
Map the straight line joining $\mathbf{A(-2+j)}$ and $\mathbf{B(3+j6)}$ in the z-plane on to the w-plane when $\mathbf{w = u + jv = f(z) = 3 + j2z}$

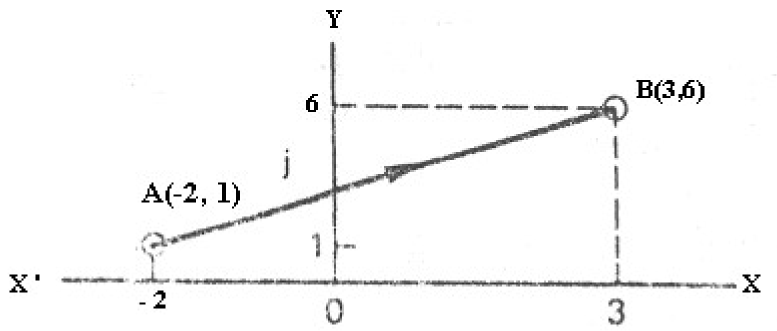

Figure 09

Answer:
We have,

$$w = f(z) = 3 + j2z$$
$$u + jv = f(z) = 3 + j2(x + jy) \qquad [w = u + jv;\ z = x + jy]$$
$$u + jv = f(z) = 3 + j2x + j^2 2y$$
$$u + jv = f(z) = 3 + j2x - 2y [\because j^2 = -1]$$
$$u + jv = f(z) = 3 + j2x - 2y$$
$$u + jv = f(z) = 3 - 2y + j2x$$
$$u + jv = f(z) = (3 - 2y) + j2x \text{ --------------------(i)}$$

Equating real and imaginary part, we get

$\mathbf{u = 3 - 2y}$ ----------------------------------(ii)

$\mathbf{v = 2x}$ ----------------------------------(iii)

Given, the point A, $\mathbf{z = -2 + j.1}$
That is A (-2, 1) ----------------------------------(iv)
Here, $\mathbf{x = -2,\ y = 1}$
Putting the value of x and y in (ii) and (iii),
Then the point A, $\mathbf{(z = -2 + j)}$ transforms onto w-plane is

$\mathbf{u = 3 - 2y}$ and $\mathbf{v = 2x}$
$\mathbf{u = 3 - 2.1}$ $\qquad \mathbf{v = 2.(-2)}$
$\mathbf{u = 1}$ $\qquad \mathbf{v = -4}$

$\therefore$ **The image of A is** $\mathbf{A'(w = u + jv = 1 - 4j)}$

That is $A'(1,-4)$ **----------------------------------(v)**

Again,
Given, the point B, $\mathbf{z = 3 + j6}$
That is B (3, 6) ----------------------------------(vi)
Here, $\mathbf{x = 3,\ y = 6}$
Putting the value of x and y in (ii) and (iii),

Then the point B, $\mathbf{z = 3 + j6}$ transforms onto w-plane is

$$\mathbf{u = 3 - 2y} \qquad \text{and} \qquad \mathbf{v = 2x}$$
$$\mathbf{u = 3 - 2.6} \qquad \mathbf{v = 2.3}$$
$$\mathbf{u = -9} \qquad \mathbf{v = 6}$$

$\therefore$ **The image of B is** $\mathbf{B'(w = u + jv = -9 + 6j)}$

That is $\boldsymbol{B'(-9,6)}$ ---------------------------------**(vii)**

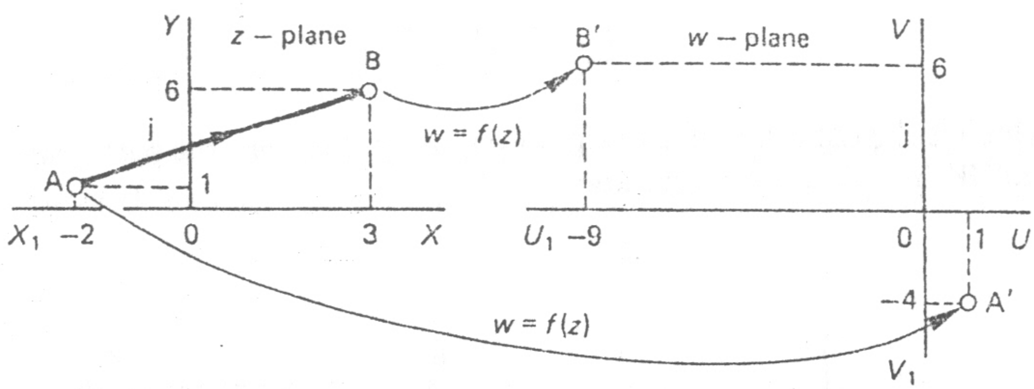

Figure 10

Again,

From Figure 09:

Given, the point A, $\mathbf{z = -2 + j.1}$

That is A (-2, 1)

Here, $\boldsymbol{x_1 = -2,\ y_1 = 1}$

Given, the point B, $\mathbf{z = 3 + j6}$

That is, B (3, 6),

Here, $\boldsymbol{x_2 = 3,\ y_2 = 6}$

The equation of the straight line AB is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 1}{1 - 6} = \frac{x - (-2)}{-2 - 3}$$

$$\Rightarrow \frac{y - 1}{-5} = \frac{x + 2}{-5}$$

$$\Rightarrow y - 1 = x + 2$$

$$\Rightarrow y = x + 3 \text{---------------------------------(viii)}$$

We have, from (ii) & (iii)

$$\mathbf{u = 3 - 2y} \qquad \mathbf{v = 2x}$$
$$\Rightarrow \mathbf{3 - 2y = u} \qquad \Rightarrow \mathbf{2x = v}$$
$$\Rightarrow \mathbf{-2y = u - 3} \qquad \Rightarrow \mathbf{x = \frac{v}{2}}$$

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$$\Rightarrow 2y = -u + 3$$
$$\Rightarrow 2y = 3 - u$$
$$\Rightarrow y = \frac{3 - u}{2}$$

Putting the value of **x** and **y** in (viii),

$$\Rightarrow y = x + 3$$
$$\Rightarrow \frac{3-u}{2} = \frac{v}{2} + 3$$
$$\Rightarrow \frac{3-u}{2} = \frac{v+6}{2}$$
$$\Rightarrow 3 - u = v + 6$$
$$\Rightarrow v + 6 = 3 - u$$
$$\Rightarrow v = -6 + 3 - u$$
$$\Rightarrow v = -3 - u$$
$$\Rightarrow v = -u - 3 \text{ -------------------------------------(ix)}$$

The equation (ix) is the equation of the straight line $\mathbf{A'B'}$

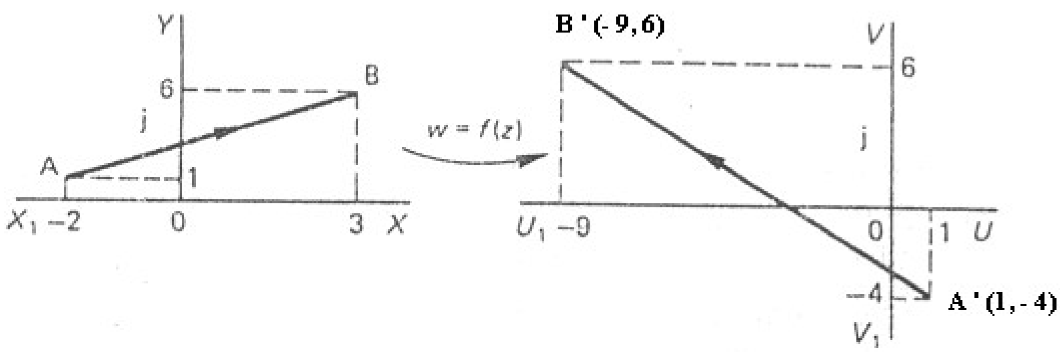

Figure 11

Justification:

We have

The image of A is $\mathbf{A'(w = u + jv = 1 - 4j)}$

That is $A'(1,-4)$

Here, $u_1 = 1,\ v_1 = -4$

and

The image of B is $\mathbf{B'(w = u + jv = -9 + 6j)}$

That is $B'(-9,6)$

Here, $u_2 = -9,\ v_2 = 6$

The equation of the straight line $\mathbf{A'B'}$ is:

$$\frac{v - v_1}{v_1 - v_2} = \frac{u - u_1}{u_1 - u_2} \qquad \left[\text{As we know: } \frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}\right]$$

$$\Rightarrow \frac{v-(-4)}{-4-6} = \frac{u-1}{1-(-9)}$$
$$\Rightarrow \frac{v+4}{-10} = \frac{u-1}{1+9}$$
$$\Rightarrow \frac{v+4}{-10} = \frac{u-1}{10}$$
$$\Rightarrow \frac{v+4}{-1} = \frac{u-1}{1}$$
$$\Rightarrow v+4 = -1(u-1)$$
$$\Rightarrow v+4 = -u+1$$
$$\Rightarrow v = -u+1-4$$
$$\Rightarrow v = -u-3 \text{------------------------------------}(x)$$

Since the equation (ix) and (x) is same. ***Hence proved***

Example 04: Graph of Parabola

$$\mathbf{y^2 = 4ax}$$

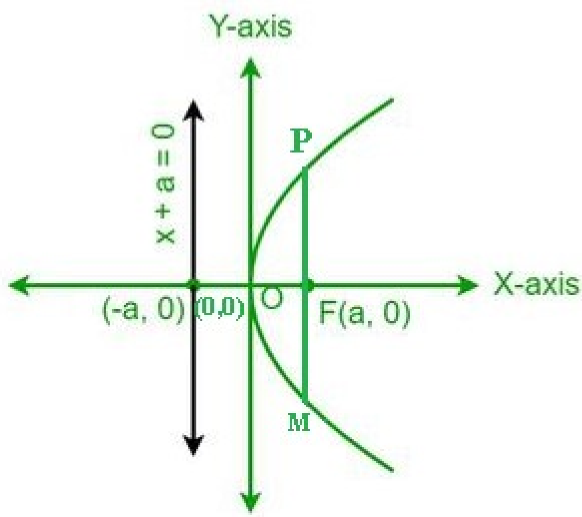

Figure 12

Focus F (a, 0)
PM= Latus Rectum = 4a

Example 05: Graph of Parabola

$$y^2 = -4ax$$

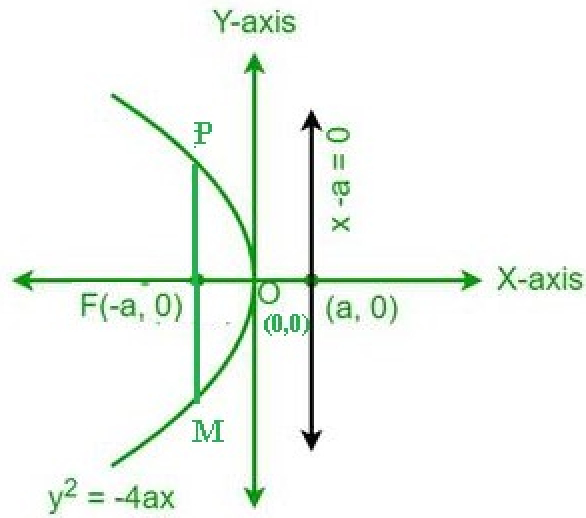

Figure 13

Focus F (- a, 0)
PM= Latus Rectum = 4a

Example 06: Graph of Parabola

$$x^2 = 4ay$$

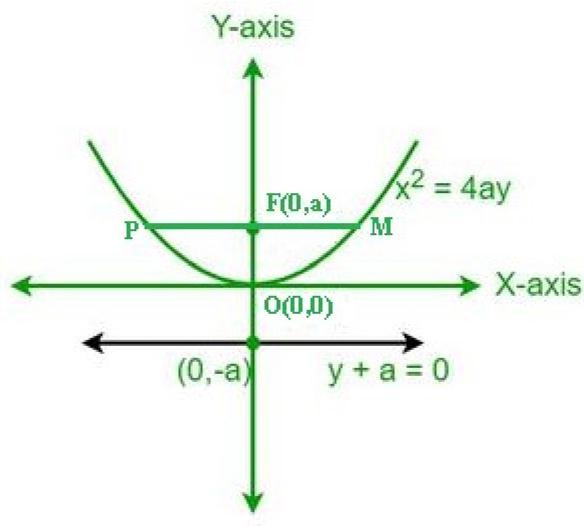

Figure 14

Focus F (0, a)
PM= Latus Rectum = 4a

Example 07: Graph of Parabola

$$x^2 = -4ay$$

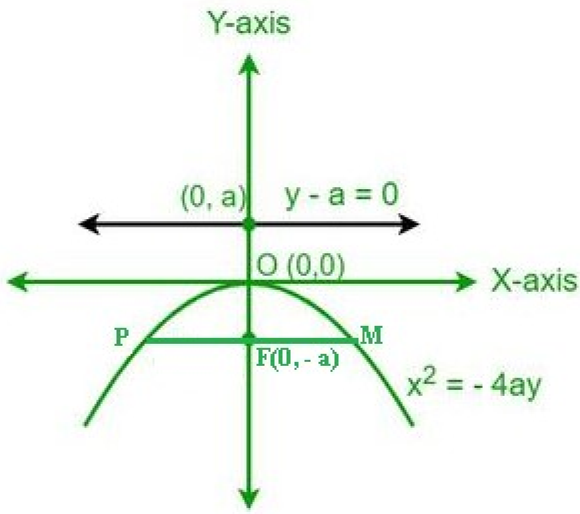

Figure 15

Focus F (0, -a)
PM= Latus Rectum = 4a

Example 08

A straight line joining $A(2+0.j)$ and $B(0+2j)$ in the z-plane is mapped onto the w-plane by the transformation equation $w = z^2$,

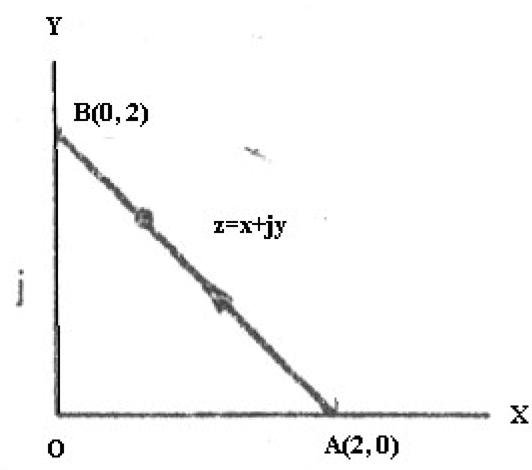

Figure 16

Solution:

We have, $w = f(z) = z^2$

$$u + jv = f(z) = (x + jy)^2 \qquad [w = u + jv;\ z = x + jy]$$

$$u + jv = x^2 + j2xy + j^2y^2$$

$$u + jv = x^2 + j2xy - y^2 \qquad [\because j^2 = -1]$$

$$u + jv = x^2 - y^2 + j2xy \text{ ---------------(i)}$$

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Equating the coefficient of real and imaginary part, we get

$\therefore u = x^2 - y^2$ ----------------- *(ii)*

$\therefore v = 2xy$ ----------------- *(iii)*

Given, $\mathbf{A(2+j.0)}$

That is, $A(2,0)$ -----------------------------(iv)

Here x = 2, y = 0

Putting the values of x and y in (ii) and (iii),

$$\mathbf{u = x^2 - y^2} \qquad \mathbf{v = 2xy}$$
$$\mathbf{u = 2^2 - 0^2} \qquad \mathbf{v = 2.2.0}$$
$$\mathbf{u = 4 - 0} \qquad \mathbf{v = 0}$$
$$\mathbf{u = 4}$$

$\therefore \mathbf{w = u + jv = 4 + j.0}$

The image of A is $\mathbf{A'(w = 4 + j.0)}$

That is $A'(4,0)$ -----------------------------(v)

Again,

$\mathbf{B(z = 0 + j2)}$

That is, $B(0,2)$ -----------------------------(vi)

Putting the values of x and y in (ii) and (iii),

$$\mathbf{u = x^2 - y^2} \qquad \mathbf{v = 2xy}$$
$$\mathbf{u = 0^2 - 2^2} \qquad \mathbf{v = 2.0.2}$$
$$\mathbf{u = 0 - 4} \qquad \mathbf{v = 0}$$
$$\mathbf{u = -4}$$
$$\therefore \mathbf{w = u + jv = -4 + j.0}$$

The image of B is $B'(w = u + jv = -4 + j.0)$

That is $B'(-4,0)$ -----------------------------(vii)

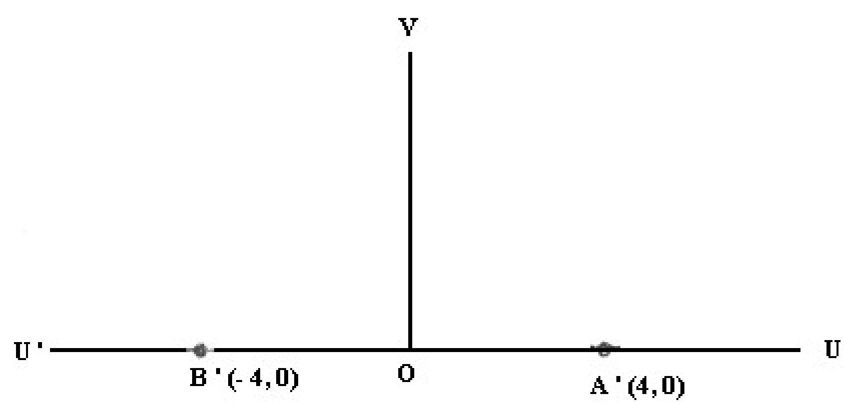

Figure 17

Given, From Figure 16:
We have: A (2, 0) and B (0, 2)

$$A : x_1 = 2, y_1 = 0$$
$$B : x_2 = 0, y_2 = 2$$

The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$
$$\Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 2}{2 - 0}$$
$$\Rightarrow \frac{y}{-2} = \frac{x - 2}{2}$$
$$\Rightarrow 2y = -2(x - 2)$$
$$\Rightarrow y = \frac{-2(x - 2)}{2}$$
$$\therefore y = -(x - 2)$$
$$\therefore y = -x + 2$$
$$\therefore y = 2 - x \qquad \text{-----------------------------(viii)}$$

Putting the value of y in (ii),

$$\therefore u = x^2 - y^2$$
$$\Rightarrow u = x^2 - (2 - x)^2$$
$$\Rightarrow u = x^2 - (4 - 4x + x^2)$$
$$\Rightarrow u = x^2 - 4 + 4x - x^2$$
$$\Rightarrow u = 4x - 4$$
$$\Rightarrow u + 4 = 4x$$
$$\Rightarrow x = \frac{u + 4}{4} \qquad \text{------------------------------(ix)}$$

Putting the value of y in (iii),

$$v = 2xy$$
$$v = 2x(2 - x)$$
$$\therefore v = 4x - 2x^2 \qquad \text{--------------------------------------(x)}$$

Now putting the value of x in equation (x)

$$\therefore v = 4x - 2x^2$$
$$v = 4\left(\frac{u + 4}{4}\right) - 2\left(\frac{u + 4}{4}\right)^2$$
$$\Rightarrow v = u + 4 - 2\left(\frac{u^2 + 8u + 16}{16}\right)$$
$$\Rightarrow v = u + 4 - \frac{1}{8}\left(u^2 + 8u + 16\right)$$

$$\Rightarrow v = u + 4 - \frac{1}{8}u^2 - u - 2$$

$$\Rightarrow v = 2 - \frac{1}{8}u^2$$

$$\Rightarrow v = \frac{16 - u^2}{8}$$

$$\Rightarrow v = -\frac{1}{8}(u^2 - 16)$$

$$\Rightarrow 8v = -(u^2 - 16)$$

$$\Rightarrow 8v = -u^2 + 16$$

$$\Rightarrow 8v = -u^2 + 16$$

$$\Rightarrow -u^2 = 8v - 16$$

$$\Rightarrow u^2 = -8(v - 2)$$

$$\Rightarrow u^2 = -4.2(v - 2) \text{ -----------------------------(xi)}$$

The equation (xi) represents an equation of a parabola.

Let,

$$U = u \;\; and \;\; V = v - 2 \text{ ------------------------------(xii)}$$

From (xii),
When,

$$U = 0 \text{ then } u = 0$$

and

$$V = 0 \;\text{ then}$$

$$\Rightarrow 0 = v - 2$$

$$\Rightarrow v = 2$$

$$\therefore \text{Vertex} = (u, v) = (0,2)$$

And latus rectum $A'B' =$

$$= 4a$$

$$= 4 \cdot 2$$

$$= 8$$

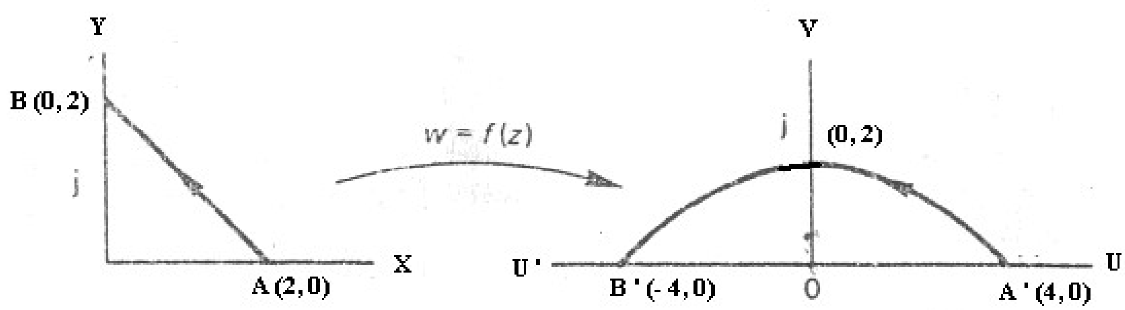

Figure 18

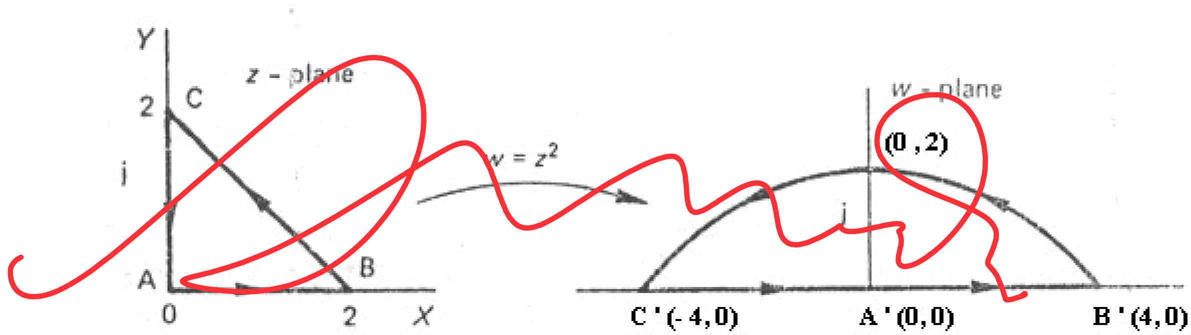

Figure 26

Example 11
A straight line joining $\mathbf{A(-j)}$ and $\mathbf{B(2+j)}$ in the z-plane is mapped onto the w-plane by the transformation equation $\mathbf{w = \dfrac{1}{z}}$

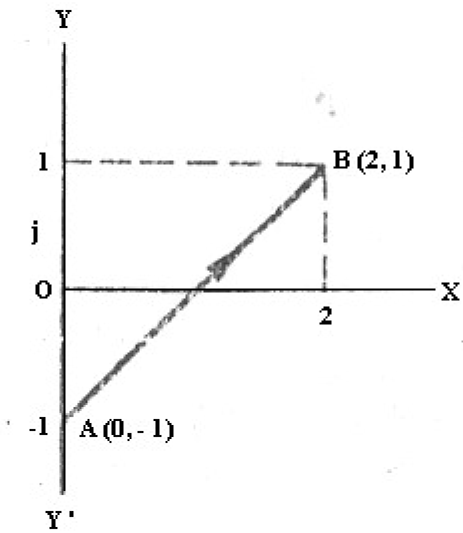

Figure 27

Solution:
Given,

$$w = \frac{1}{z}$$

$$w = \frac{1}{x + jy} \qquad [z = x + jy]$$

$$w = \frac{x - jy}{(x + jy)(x - jy)} \qquad \text{[Multiplying by } x - jy\text{]}$$

$$w = \frac{x - jy}{x^2 - jxy + jxy - j^2 y^2}$$

$$w = \frac{x - jy}{x^2 + y^2} \qquad [\because j^2 = -1]$$

$$u + jv = \frac{x - jy}{x^2 + y^2} \qquad [w = u + jv]$$

$$u + jv = \frac{x}{x^2 + y^2} - j\frac{y}{x^2 + y^2} \qquad \text{------------------------(i)}$$

Equating the coefficient of real and imaginary part, we get,

$$u = \frac{x}{x^2 + y^2} \qquad ------------(ii)$$

$$v = \frac{-y}{x^2 + y^2} \qquad ----------(iii)$$

Given, $\mathbf{A(0 - j.1)}$

That is, $A(0,-1)$ ------------------------(iv)

Here,

$\mathbf{x = 0, y = -1}$

Putting the value of x and y in (ii) and (iii)

$$\mathbf{u = \frac{x}{x^2 + y^2}} \qquad \mathbf{v = \frac{-y}{x^2 + y^2}}$$

$$\mathbf{u = \frac{0}{0 + (-1)^2}} \qquad \mathbf{v = \frac{-(-1)}{0^2 + (-1)^2}}$$

$$\mathbf{u = \frac{0}{0 + 1}} \qquad \mathbf{v = \frac{1}{1}}$$

$$\mathbf{u = \frac{0}{1}} \qquad \mathbf{v = 1}$$

$$\mathbf{u = 0} \qquad \mathbf{v = 1}$$

$$\therefore \mathbf{w = u + jv = 0 + j.1}$$

The image of A is $\mathbf{A'(w = 0 + j.1)}$

That is $A'(0,1)$ ------------------------(v)

Again,

$$\mathbf{B(z = 2 + j.1)}$$

That is, $B(2,1)$ ------------------------(vi)

Here, $\mathbf{x = 2, y = 1}$

Putting the value of x and y in (ii) and (iii),

$$\mathbf{u = \frac{x}{x^2 + y^2}} \qquad \mathbf{v = \frac{-y}{x^2 + y^2}}$$

$$\mathbf{u = \frac{2}{2^2 + 1^2}} \qquad \mathbf{v = \frac{-1}{2^2 + 1^2}}$$

$$\mathbf{u = \frac{2}{5}} \qquad \mathbf{v = \frac{-1}{5}}$$

$$\therefore w = u + jv = \frac{2}{5} - j\frac{1}{5}$$

The image of B is $B'(\, w = \frac{2}{5} - j\frac{1}{5})$

That is $B'(\frac{2}{5}, -\frac{1}{5})$ ------------------------(vii)

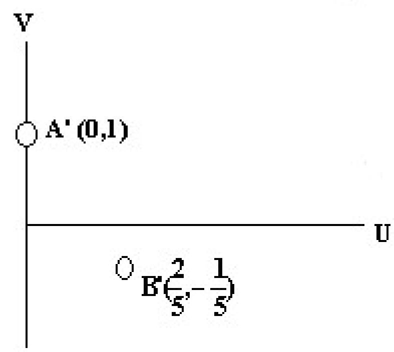

Figure 28

From Figure 27:
Given A(0,-1) and B(2,1)
The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - (-1)}{-1 - 1} = \frac{x - 0}{0 - 2}$$

$$\Rightarrow \frac{y + 1}{-1 - 1} = \frac{x - 0}{0 - 2}$$

$$\Rightarrow \frac{y + 1}{-2} = \frac{x}{-2}$$

$$\Rightarrow y + 1 = x$$

$$\therefore y = x - 1 \qquad ----------(viii)$$

Again, Given

$$w = \frac{1}{z}$$

$$\therefore z = \frac{1}{w}$$

$$z = \frac{1}{u + jv} \qquad [w = u + jv]$$

$$z = \frac{u - jv}{(u + jv)(u - jv)}$$

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$$z=\frac{u-jv}{u^2-(jv)^2}$$

$$z=\frac{u-jv}{u^2+v^2} \qquad [\because j^2=-1]$$

$$x+jy=\frac{u-jv}{u^2+v^2} \qquad [z=x+jy]$$

$$i.e.\ x+jy=\frac{u}{u^2+v^2}-j\frac{v}{u^2+v^2}--------(ix)$$

Equating the coefficient of real and imaginary part, we get,

$$x=\frac{u}{u^2+v^2}\ ; \qquad y=\frac{-v}{u^2+v^2}\ ---------(x)$$

Putting the value of x and y in (viii),

$$\mathbf{y=x-1}$$

$$\Rightarrow \frac{-v}{u^2+v^2}=\frac{u}{u^2+v^2}-1$$

$$\Rightarrow \frac{-v}{u^2+v^2}=\frac{u-u^2-v^2}{u^2+v^2}$$

$$\Rightarrow -v=u-u^2-v^2$$

$$\Rightarrow u-u^2-v^2+v=0$$

$$\Rightarrow -u+u^2+v^2-v=0$$

$$\Rightarrow u^2-u+v^2-v=0$$

$$\Rightarrow (u^2-u)+(v^2-v)=0$$

$$\Rightarrow u^2-2\cdot u\cdot\frac{1}{2}+\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2+v^2-2\cdot v\cdot\frac{1}{2}+\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2=0$$

$$\Rightarrow u^2-2\cdot u\cdot\frac{1}{2}+\left(\frac{1}{2}\right)^2+v^2-2\cdot v\cdot\frac{1}{2}+\left(\frac{1}{2}\right)^2-\frac{1}{4}-\frac{1}{4}=0$$

$$\Rightarrow \left(u-\frac{1}{2}\right)^2+\left(v-\frac{1}{2}\right)^2-\frac{2}{4}=0$$

$$\Rightarrow \left(u-\frac{1}{2}\right)^2+\left(v-\frac{1}{2}\right)^2-\frac{1}{2}=0$$

$$\Rightarrow \left(u-\frac{1}{2}\right)^2+\left(v-\frac{1}{2}\right)^2=\frac{1}{2}$$

$$\Rightarrow \left(u-\frac{1}{2}\right)^2+\left(v-\frac{1}{2}\right)^2=\left(\frac{1}{\sqrt{2}}\right)^2----------(xi)$$

The equation (xi) represents an equation of a circle whose centre $C\left(\frac{1}{2},\frac{1}{2}\right)$ and

$\mathbf{Radius} = \frac{1}{\sqrt{2}}$

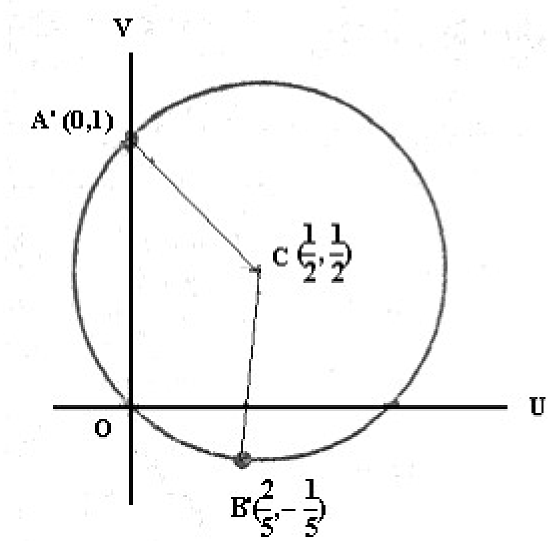


Figure 29

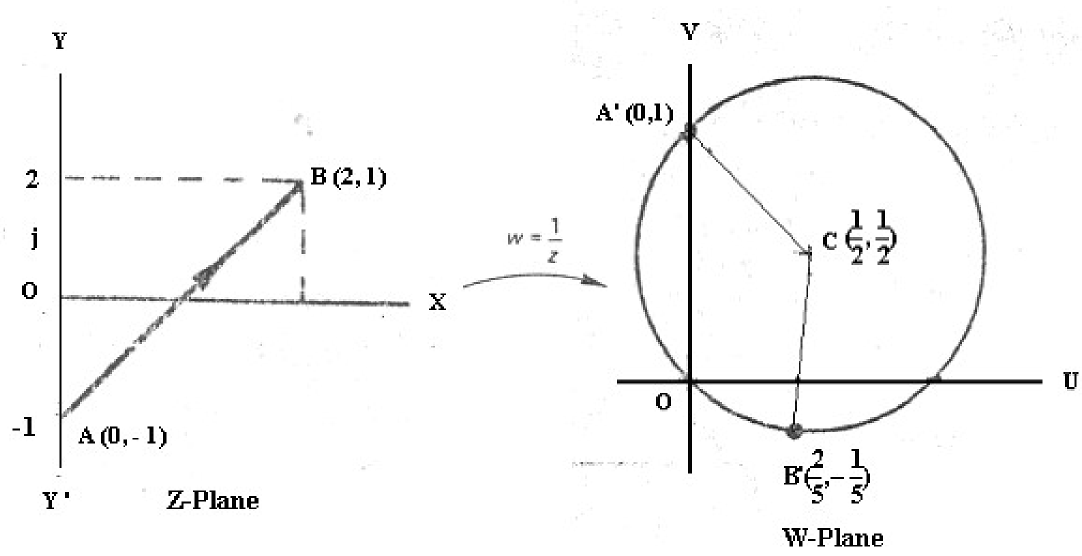


Figure 30

Justification of radius of the circle:

We have $\mathbf{A'(w = 0 + j.1)}$ that is the coordinate of $\mathbf{A'(0,1)}$ and the center $\mathbf{C}\left(\frac{1}{2},\frac{1}{2}\right)$

$$\therefore A'C = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$\therefore \mathbf{A'C} = \sqrt{(0 - \frac{1}{2})^2 + (1 - \frac{1}{2})^2}$$

$$\therefore \mathbf{A'C} = \sqrt{(\frac{1}{2})^2 + (\frac{1}{2})^2}$$

$$\therefore \mathbf{A'C} = \sqrt{\frac{1}{4} + \frac{1}{4}}$$

$$\therefore \mathbf{A'C} = \sqrt{\frac{2}{4}}$$

$$\therefore \mathbf{A'C} = \sqrt{\frac{1}{2}}$$

$$\therefore A'C = \frac{1}{\sqrt{2}} \text{ (Proved)}$$

$$\therefore \text{Radius} = \frac{1}{\sqrt{2}}$$

We have $B'(\frac{2}{5}, -\frac{1}{5})$ and $C(\frac{1}{2}, \frac{1}{2})$

$$\therefore B'C = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$B'C = \sqrt{(\frac{2}{5} - \frac{1}{2})^2 + (-\frac{1}{5} - \frac{1}{2})^2}$$

$$B'C = \sqrt{(\frac{4-5}{10})^2 + (\frac{-2-5}{10})^2}$$

$$B'C = \sqrt{(\frac{-1}{10})^2 + (\frac{-7}{10})^2}$$

$$B'C = \sqrt{\frac{1}{100} + \frac{49}{100}}$$

$$B'C = \sqrt{\frac{50}{100}}$$

$$B'C = \sqrt{\frac{1}{2}}$$

$$\therefore B'C = \frac{1}{\sqrt{2}} \text{ (Proved)}$$

$$\therefore \text{Radius} = \frac{1}{\sqrt{2}}$$

Example 12

A circle in the z-plane has its centre at $\mathbf{z = 3}$ and a radius of 2 units. Determine its image in the w-plane when transformation by $\mathbf{w = \frac{1}{z}}$

Where c is the circle $|z-3| = 2$

We have,

$z = x + jy$

$z - 3 = x + jy - 3$

$z - 3 = x - 3 + jy$

$\therefore |z-3| = \sqrt{(x-3)^2 + y^2}$

Given,

$|z-3| = 2$

$\therefore |z-3| = \sqrt{(x-3)^2 + y^2} = 2$

$\therefore \sqrt{(x-3)^2 + y^2} = 2$

$\therefore (x-3)^2 + y^2 = 2^2$

$\therefore (x-3)^2 + (y-0)^2 = 2^2$

This is the equation of the circle whose centre (3,0) and radius 2

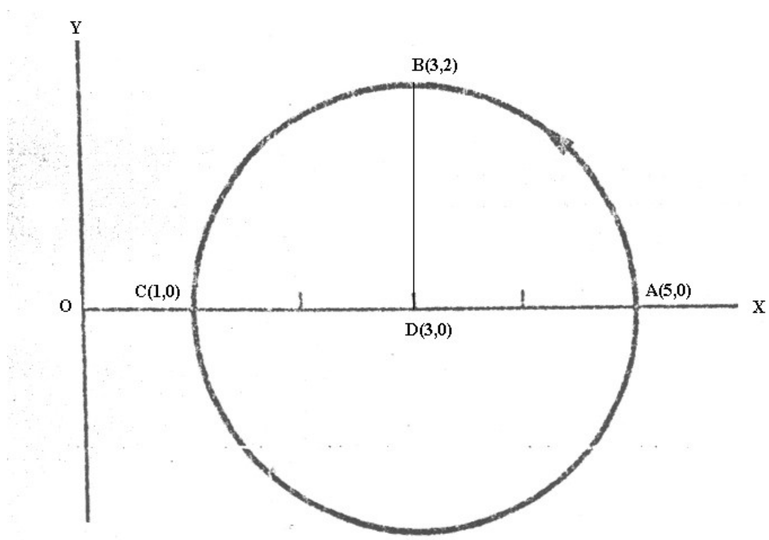


Figure 31

Solution:

Given,

$$\mathbf{z = 3}$$
$$\Rightarrow \mathbf{x + jy = 3}$$
$$\Rightarrow \boldsymbol{x + jy = 3 + 0.j} \text{ -------------------------(i)}$$

Equating the coefficient of real and imaginary part, we get,

$$x = 3, \qquad y = 0$$

Hence, we can write,

$$(x, y) = (3,0)$$

and given, radius =2

So, Equation of the circle

$(x-3)^2 + (y-0)^2 = 2^2$ ------------------------(ii)

$$[\because (x-a)^2 + (y-b)^2 = r^2]$$

That is centre of the circle is (3,0) and radius 2

$$(x-3)^2 + y^2 = 4$$

$$\Rightarrow x^2 - 6x + 9 + y^2 = 4$$

$$\Rightarrow x^2 + y^2 - 6x + 5 = 0 \qquad ------------(iii)$$

Again, Given,

$$w = \frac{1}{z}$$

$$w = \frac{1}{x + jy} \qquad [z = x + jy]$$

$$w = \frac{x - jy}{(x + jy)(x - jy)} \qquad \text{[Multiplying by } x - jy\text{]}$$

$$w = \frac{x - jy}{x^2 - jxy + jxy - j^2 y^2}$$

$$w = \frac{x - jy}{x^2 + y^2} \qquad [\because j^2 = -1]$$

$$u + jv = \frac{x - jy}{x^2 + y^2} \qquad [w = u + jv]$$

$$u + jv = \frac{x}{x^2 + y^2} - j\frac{y}{x^2 + y^2} \qquad ------------------------(iv)$$

Equating the coefficient of real and imaginary part, we get,

$$u = \frac{x}{x^2 + y^2} \qquad -----------(v)$$

$$v = \frac{-y}{x^2 + y^2} \qquad ----------(vi)$$

Again, Given

$$w = \frac{1}{z}$$

$$\therefore z = \frac{1}{w}$$

$$z = \frac{1}{u + jv} \qquad [w = u + jv]$$

$$z=\frac{u-jv}{(u+jv)(u-jv)}$$

$$z=\frac{u-jv}{u^2-(jv)^2}$$

$$z=\frac{u-jv}{u^2+v^2} \qquad [\because j^2=-1]$$

$$x+jy=\frac{u-jv}{u^2+v^2} \qquad [z=x+jy]$$

$$\text{i.e. } x+jy=\frac{u}{u^2+v^2}-j\frac{v}{u^2+v^2} \text{---------}(vii)$$

Equating the coefficient of real and imaginary part, we get,

$$x=\frac{u}{u^2+v^2}\ ; \qquad y=\frac{-v}{u^2+v^2} \text{----------}(viii)$$

Substituting the values of x and y in (iii),

$$x^2+y^2-6x+5=0$$

$$\Rightarrow\left(\frac{u}{u^2+v^2}\right)^2+\left(\frac{-v}{u^2+v^2}\right)^2-6\left(\frac{u}{u^2+v^2}\right)+5=0$$

$$\Rightarrow\frac{u^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}-\frac{6u}{u^2+v^2}+5=0$$

$$\Rightarrow\frac{u^2+v^2}{(u^2+v^2)^2}-\frac{6u}{u^2+v^2}+5=0$$

$$\Rightarrow\frac{1}{u^2+v^2}-\frac{6u}{u^2+v^2}+5=0$$

$$\Rightarrow\frac{1-6u+5(u^2+v^2)}{u^2+v^2}=0$$

$$\Rightarrow 5(u^2+v^2)-6u+1=0$$

$$\Rightarrow u^2+v^2-\frac{6}{5}u+\frac{1}{5}=0 \ \text{[dividing by 5]}$$

$$\Rightarrow u^2+v^2-2\cdot\frac{3}{5}\cdot u+2\cdot 0\cdot v+\frac{1}{5}=0$$

$$\Rightarrow u^2+v^2+2\cdot(-\frac{3}{5})\cdot u+2\cdot 0\cdot v+\frac{1}{5}=0 \text{------------}(ix)$$

We know the general equation of the circle is

$$x^2+y^2+2gx+2fy+c=0$$

Whose centre is $(-g,-f)$ and radius is $\sqrt{g^2+f^2-c}$

Hence from (ix)

Here $\mathbf{g} = -\dfrac{3}{5}$, $\mathbf{f} = \mathbf{0}$ and $\mathbf{c} = \dfrac{1}{5}$

The centre of the new circle of (ix) is $\mathbf{(-g,-f) = (-(-\tfrac{3}{5}),-0) = (\tfrac{3}{5},0)}$

That is, centre of new circle in the w-plane is, $D(\dfrac{3}{5},0)$ [From figure 32]

Radius is $\sqrt{g^2+f^2-c} = \sqrt{(-\dfrac{3}{5})^2 + 0^2 - \dfrac{1}{5}}$

$$= \sqrt{(-\frac{3}{5})^2 + 0^2 - \frac{1}{5}} = \sqrt{\frac{9}{25} + 0 - \frac{1}{5}}$$

$$= \sqrt{\frac{9}{25} - \frac{1}{5}}$$

$$= \sqrt{\frac{9-5}{25}}$$

$$= \sqrt{\frac{4}{25}}$$

$$= \frac{2}{5}$$

That is, radius of new circle in the w-plane is, $\dfrac{2}{5}$

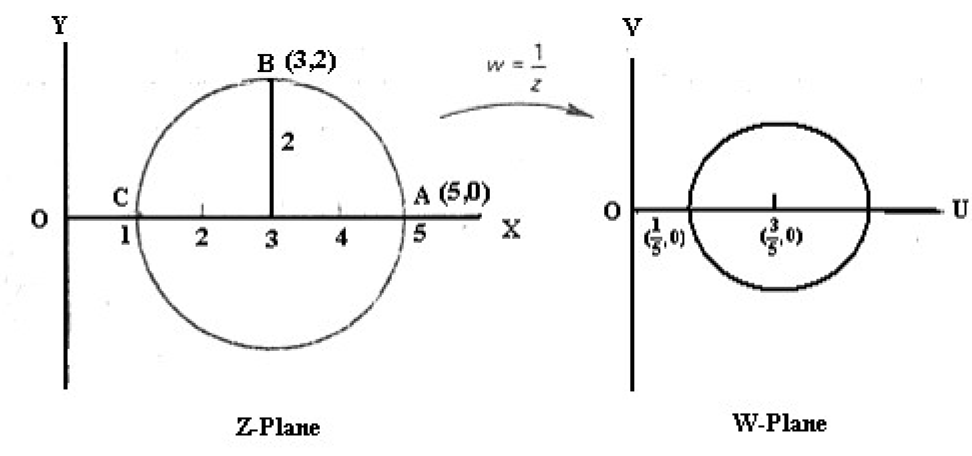


Figure 32

Taking three sample points A, B, C as shown, that is:
A(5,0), **B(3,2)**, **C(1,0)**

Putting the values of **A(5,0)**, **B(3,2)** **C(1,0)** in (v) and (vi)

We have,

$$u = \frac{x}{x^2 + y^2} \qquad v = \frac{-y}{x^2 + y^2}$$

For $\mathbf{A(5,0)}$; $u = \frac{x}{x^2 + y^2} = \frac{5}{5^2 + 0^2} = \frac{5}{25 + 0} = \frac{5}{25} = \frac{1}{5}$

For $\mathbf{A(5,0)}$; $v = \frac{-y}{x^2 + y^2} = \frac{-0}{5^2 + 0^2} = \frac{0}{25 + 0} = 0$

$\therefore$ For $\mathbf{A(5,0)}$; $w = u + jv = \frac{1}{5} + j.0$

The image of A is $\mathbf{A'(w = u + jv = \frac{1}{5} + j.0) = \frac{1}{5} + j.0}$

That is $A'(\frac{1}{5},0)$ ---------------------(x)

For $\mathbf{B(3,2)}$; $u = \frac{x}{x^2 + y^2} = \frac{3}{3^2 + 2^2} = \frac{3}{9 + 4} = \frac{3}{13}$

For $\mathbf{B(3,2)}$; $v = \frac{-y}{x^2 + y^2} = \frac{-2}{3^2 + 2^2} = \frac{-2}{9 + 4} = \frac{-2}{13}$

$\therefore$ For $\mathbf{B(3,2)}$; $w = u + jv = \frac{3}{13} + j.(\frac{-2}{13})$

The image of B is $\mathbf{B'(w = u + jv = \frac{3}{13} + j.(-\frac{2}{13})) = \frac{3}{13} - j.\frac{2}{13}}$

That is $B'(\frac{3}{13},-.\frac{2}{13})$ -------------------------(xi)

For $\mathbf{C(1,0)}$; $u = \frac{x}{x^2 + y^2} = \frac{1}{1^2 + 0^2} = \frac{1}{1} = 1$

For $\mathbf{C(1,0)}$; $v = \frac{-y}{x^2 + y^2} = \frac{-0}{1^2 + 0^2} = \frac{-0}{1} = 0$

$\therefore$ For $\mathbf{C(1,0)}$; $w = u + jv = 1 + j.(0)$

The image of C is $\mathbf{C'(w = u + jv = 1 + j.0 = 1 + j.0}$

That is $C'(1,0)$ --------------------------(xii)

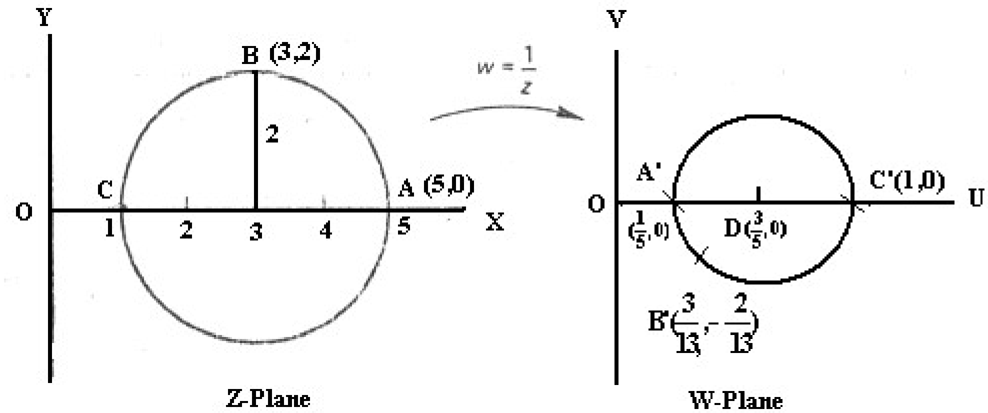


Figure 33

<u>Justification:</u>

We have $A'(\frac{1}{5},0), B'(\frac{3}{13},-\frac{2}{13}), C'(1,0), D(\frac{3}{5},0)$

Radius $= \frac{2}{5}$

We know the length of a line between two points (x_1, y_1) and (x_2, y_2) is:

$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Here, in the w-plane

$\therefore A'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$

$\therefore A'D = \sqrt{(\frac{1}{5} - \frac{3}{5})^2 + (0-0)^2}$

$\therefore A'D = \sqrt{(-\frac{2}{5})^2}$

$\therefore A'D = \sqrt{\frac{4}{25}}$

$\therefore A'D = \frac{2}{5}$ (Proved)

Again,

$\therefore B'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$

$$B'D = \sqrt{(\frac{3}{13} - \frac{3}{5})^2 + (\frac{-2}{13} - 0)^2}$$

$$B'D = \sqrt{(\frac{15-39}{65})^2 + (\frac{4}{169})}$$

$$B'D = \sqrt{(\frac{-24}{65})^2 + \frac{4}{169}}$$

$$B'D = \sqrt{\frac{576}{4225} + \frac{4}{169}}$$

$$B'D = \sqrt{\frac{114244}{714025}}$$

$$B'D = \frac{2}{5} \qquad \text{(Proved)}$$

$$\therefore C'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$C'D = \sqrt{(1 - \frac{3}{5})^2 + (0 - 0)^2}$$

$$C'D = \sqrt{(\frac{5-3}{5})^2}$$

$$C'D = \sqrt{(\frac{2}{5})^2}$$

$$C'D = \frac{2}{5} \qquad \text{(Proved)}$$

Example 13

A circle $|z| = 1$ in the Z-plane is mapped onto the W-plane by $\mathbf{w} = \frac{1}{\mathbf{z - 2}}$

Solution: from figure 34

$OP = |z| = \sqrt{x^2 + y^2}$

Given,

$|z| = 1$

$\sqrt{x^2 + y^2} = 1$

$\therefore x^2 + y^2 = 1$

$(x - 0)^2 + (y - 0)^2 = 1^2$ ----------------(i)

[We have, $(x - a)^2 + (y - b)^2 = r^2$]

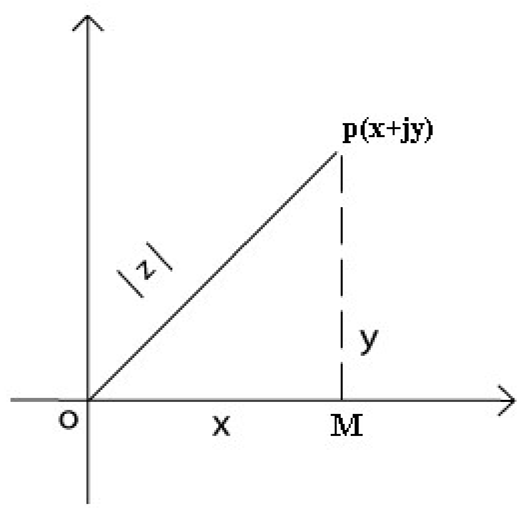
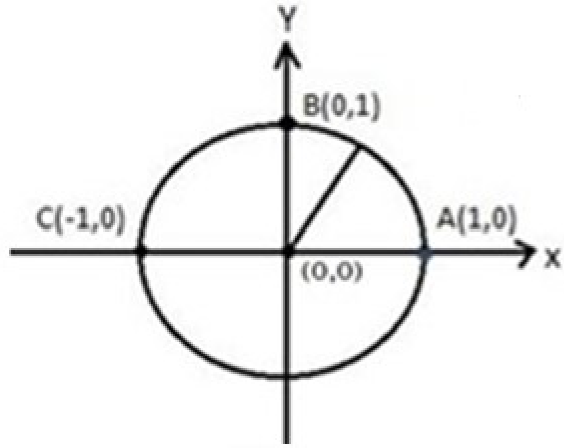


Figure 34

Which is the equation of a circle whose Center (0, 0), Radius=1



Z-Plane

Figure 35

From (i)

$$(x-0)^2+(y-0)^2=1^2$$

$$x^2+y^2=1$$

$$x^2+y^2-1=0 \qquad \text{-------------------------------(ii)}$$

Given,

$$w=\frac{1}{z-2}$$

$$z-2=\frac{1}{w}$$

$$\therefore z=\frac{1}{w}+2$$

i.e.

$$x+jy=\frac{1}{w}+2 \qquad [z=x+jy]$$

$$x+jy=\frac{1}{u+jv}+2 \qquad [w=u+jv]$$

$$x+jy=\frac{u-jv}{(u+jv)(u-jv)}+2 \qquad \text{[Multiplying by } u-jv\text{]}$$

$$x+jy=\frac{u-jv}{u^2-(jv)^2}+2$$

$$x+jy=\frac{u-jv}{u^2-j^2v^2}+2$$

$$x+jy=\frac{u-jv}{u^2+v^2}+2 \qquad [\,j^2=-1\,]$$

$$\therefore x-2+jy=\frac{u-jv}{u^2+v^2}$$

$$\therefore x-2+jy=\frac{u}{u^2+v^2}-j\frac{v}{u^2+v^2}$$ -------------------------(iii)

Equating the co-efficient of real and imaginary part on both sides, we get,

$$x-2=\frac{u}{u^2+v^2};\ y=\frac{-v}{u^2+v^2}$$

$$x=\frac{u}{u^2+v^2}+2\ ;\ y=\frac{-v}{u^2+v^2}$$ ------------------------------(iv)

Substituting these values x and y in equation (ii);

$$x^2+y^2-1=0$$

$$\left(\frac{u}{u^2+v^2}+2\right)^2+\left(\frac{-v}{u^2+v^2}\right)^2-1=0$$

$$\left\{\frac{u+2(u^2+v^2)}{u^2+v^2}\right\}^2+\frac{v^2}{(u^2+v^2)^2}=1$$

$$\frac{\{u+2(u^2+v^2)\}^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}=1$$

$$\frac{\{u+2(u^2+v^2)\}^2+v^2}{(u^2+v^2)^2}=1$$

$$\{u+2(u^2+v^2)\}^2+v^2=(u^2+v^2)^2$$

$$u^2+2\times u\times 2(u^2+v^2)+\{2(u^2+v^2)\}^2+v^2=(u^2+v^2)^2$$

$$u^2+4u(u^2+v^2)+\{2(u^2+v^2)\}^2+v^2=(u^2+v^2)^2$$

$$u^2+v^2+4u(u^2+v^2)+\{2(u^2+v^2)\}^2=(u^2+v^2)^2$$

$$(u^2+v^2)+4u(u^2+v^2)+\{2(u^2+v^2)\}^2=(u^2+v^2)^2$$

$$(u^2+v^2)+4u(u^2+v^2)+4(u^2+v^2)^2=(u^2+v^2)^2$$

$$1+4u+4(u^2+v^2)=u^2+v^2$$ [Dividing by u^2+v^2]

$$1+4u+4(u^2+v^2)-(u^2+v^2)=0$$

$$1+4u+3(u^2+v^2)=0$$

$$3(u^2+v^2)+4u+1=0$$

$$(u^2+v^2)+\frac{4}{3}u+\frac{1}{3}=0$$

$$u^2+\frac{4}{3}u+v^2+\frac{1}{3}=0$$

$$u^2+\frac{4}{3}u+v^2+\frac{1}{3}=0$$

$$u^2+v^2+\frac{4}{3}u+\frac{1}{3}=0$$

$$u^2+v^2+2.\frac{2}{3}.u+2.0.v+\frac{1}{3}=0 \text{ -----------------------------(v)}$$

We know the general equation of the circle is

$$x^2+y^2+2gx+2fy+c=0$$

Whose centre is $(-g,-f)$ and radius is $\sqrt{g^2+f^2-c}$

Hence from (v)

Here $g=\frac{2}{3}$, $f=0$ and $c=\frac{1}{3}$

The centre of new circle in w-plane is $(-g,-f)=((-\frac{2}{3}),-0)=(-\frac{2}{3},0)$

Radius is $\sqrt{g^2+f^2-c}=\sqrt{(\frac{2}{3})^2+0^2-\frac{1}{3}}$

$$=\sqrt{\frac{4}{9}-\frac{1}{3}}$$

$$=\sqrt{\frac{4-3}{9}}$$

$$=\sqrt{\frac{1}{9}}$$

$$=\frac{1}{3}$$

Now, we have to find out the image point A', B' *and* C'

Taking three sample points from figure 35, $A(1,0)$, $B(0,1)$ & $C(-1,0)$

Given,

$$w=\frac{1}{z-2}$$

$$w=\frac{1}{x+jy-2} \qquad [z=x+jy]$$

$$=\frac{1}{x-2+jy}$$

$$=\frac{x-2-jy}{(x-2+jy)(x-2-jy)} \qquad \text{[Multiplying by } x-2-jy\text{]}$$

$$=\frac{x-2-jy}{(x-2)^2-(jy)^2}$$

$$=\frac{x-2-jy}{(x-2)^2+y^2} \qquad [[j^2=-1;\ a^2-b^2=(a+b)(a-b)]]$$

$$w=\frac{x-2-jy}{(x-2)^2+y^2}$$

$$w=\frac{x-2}{(x-2)^2+y^2}-j\frac{y}{(x-2)^2+y^2}$$

$$u+jv=\frac{x-2}{(x-2)^2+y^2}-j\frac{y}{(x-2)^2+y^2} \qquad [w=u+jv]$$

$$u+jv=\frac{x-2}{(x-2)^2+y^2}-j\frac{y}{(x-2)^2+y^2} \qquad \text{-----------------------(vi)}$$

Equating the co-efficient of real and imaginary part from (vi), we get,

$$u=\frac{x-2}{(x-2)^2+y^2} \qquad \text{-----------------------------(vii)}$$

$$v=\frac{-y}{(x-2)^2+y^2} \qquad \text{-----------------------------(viii)}$$

For $A(1,0)$: we get from vii & viii

$$u=\frac{x-2}{(x-2)^2+y^2}=\frac{1-2}{(1-2)^2+0^2}=-\frac{1}{1}=-1$$

$$v=\frac{-y}{(x-2)^2+y^2}=\frac{-0}{(1-2)^2+0^2}=0$$

$\therefore A'(w=u+jv=-1+j.0)$ is the image of A.

That is $A'(-1,0)$ -----------------------------(ix)

For $B(0,1)$: we get from vii & viii

$$u=\frac{x-2}{(x-2)^2+y^2}=\frac{0-2}{(0-2)^2+1^2}=-\frac{2}{5}$$

$$v=\frac{-y}{(x-2)^2+y^2}=\frac{-1}{(0-2)^2+1^2}=-\frac{1}{5}$$

$\therefore B'(w=u+jv=-\frac{2}{5}-\frac{1}{5}j)$ is the image of B.

That is $B'(-\frac{2}{5},-\frac{1}{5})$ -----------------------------(x)

For $C(-1,0)$: we get from vii & viii

$$u=\frac{x-2}{(x-2)^2+y^2}=\frac{-1-2}{(-1-2)^2+0^2}=-\frac{3}{9}=-\frac{1}{3}$$

$$v=\frac{-y}{(x-2)^2+y^2}=\frac{-0}{(-1-2)^2+0^2}=0$$

$\therefore C'(w=u+jv=-\frac{1}{3}+j.0)$ is the image of C.

That is $C'(-\frac{1}{3},0)$ -----------------------------(xi)

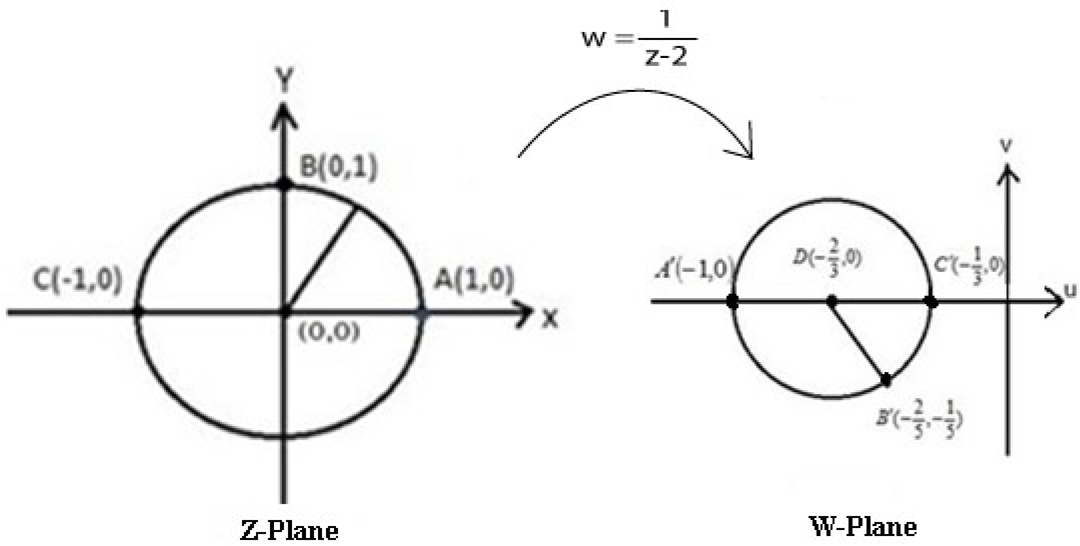


Figure 36

Justification:

We have, $A'(-1,0)\ B'(-\frac{2}{5},-\frac{1}{5})\ C'(-\frac{1}{3},0)$ and the radius & centre of the new circle (v) is Radius $=\frac{1}{3}$ and centre $D(-\frac{2}{3},0)$

We know the length of a line between two points (x_1,y_1) and (x_2,y_2) is:

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$$

Here, in the w-plane

$$\therefore A'D=\sqrt{(u_1-u_2)^2+(v_1-v_2)^2}$$

$$=\sqrt{\left(-1-(-\frac{2}{3})\right)^2+(0-0)^2}$$

$$=\sqrt{\left(-1+\frac{2}{3})\right)^2+0}=\sqrt{\left(\frac{-3+2}{3}\right)^2}$$

$$=\sqrt{\left(\frac{-1}{3}\right)^2}=\sqrt{\frac{1}{9}}=\frac{1}{3}$$

$$\therefore B'D=\sqrt{(u_1-u_2)^2+(v_1-v_2)^2}$$

$$=\sqrt{\left(-\frac{2}{5}-(-\frac{2}{3})\right)^2+\left(-\frac{1}{5}+0\right)^2}$$

$$=\sqrt{\left(-\frac{2}{5}+\frac{2}{3}\right)^2+\left(-\frac{1}{5}+0\right)^2}$$

$$= \sqrt{\left(\frac{-6+10}{15}\right)^2 + \frac{1}{25}}$$

$$= \sqrt{\frac{16}{225} + \frac{1}{25}}$$

$$= \sqrt{\frac{16+9}{225}} = \sqrt{\frac{25}{225}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\therefore C'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$= \sqrt{\left(-\frac{1}{3} - (-\frac{2}{3})\right)^2 + (0-0)^2}$$

$$= \sqrt{\left(-\frac{1}{3} + \frac{2}{3}\right)^2 + (0-0)^2}$$

$$= \sqrt{\left(\frac{-1+2}{3}\right)^2} = \sqrt{\frac{1}{9}} = \frac{1}{3}$$

Example 14: Determine the image in w-plan of circle $|z| = 2$ in the z-plane under the transformation $w = \dfrac{1}{z-2}$ and show the region in w-plan onto which the region within the circle is mapped.

Answer: from figure 37

$OP = |z| = \sqrt{x^2 + y^2}$

Given,

$|z| = 2$

$\sqrt{x^2 + y^2} = 2$

$x^2 + y^2 = 2^2$ [Squaring]

$\therefore x^2 + y^2 = 4$

$(x-0)^2 + (y-0)^2 = 2^2$ ---------------------(i)

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

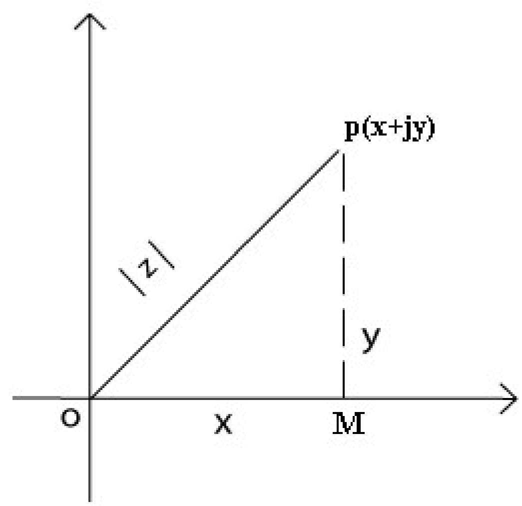


Figure no 37

The equation (i) represents a circle whose Center (0, 0) & radius = 2

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x}=\frac{\delta v}{\delta y} \quad \textit{and} \quad \frac{\delta v}{\delta x}=-\frac{\delta u}{\delta y}$$

We said earlier that where a function fails to be regular, a singular point or singularity occurs. i.e. where $\mathbf{w = f(z)}$ is not continuous or where the **Cauchy-Riemann Test fails.**

Determine where each of the following functions fails to be regular, i.e. where singularities occur.

As for example

$$\mathbf{w = f(z) = \frac{1}{(z-2)(z-3)}}$$

Singularities at $\mathbf{z = 2}$ and $\mathbf{z = 3}$ *Answer*

Example 15

Determine the function $\mathbf{w = f(z) = z^2 - 4}$ is analytic or not.

Answer:

Given,

$$\mathbf{w = f(z) = z^2 - 4}$$

$$\mathbf{\Rightarrow u + iv = (x+iy)^2 - 4} \qquad [\because \mathbf{w = u + iv \ \& \ z = x + iy}]$$

$$\mathbf{\Rightarrow u + iv = x^2 + 2xiy + i^2y^2 - 4}$$

$$\mathbf{\Rightarrow u + iv = x^2 - y^2 - 4 + 2xiy} \qquad \mathbf{[i^2 = -1]} \text{--------------------------(i)}$$

Equating real and imaginary part,

$$\mathbf{u = x^2 - y^2 - 4} \text{------------------------------------(ii)}$$

$$\mathbf{v = 2xy} \qquad \text{------------------------------------(iii)}$$

From (ii)

$$\mathbf{u = x^2 - y^2 - 4}$$

$$\mathbf{\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x^2 - y^2 - 4)}$$

$$\mathbf{\frac{\delta u}{\delta x} = 2x - 0 - 0}$$

$$\mathbf{\frac{\delta u}{\delta x} = 2x} \qquad \text{------------------------------------(iv)}$$

Again

$$\mathbf{u = x^2 - y^2 - 4}$$

$$\mathbf{\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^2 - y^2 - 4)}$$

$$\mathbf{\frac{\delta u}{\delta y} = 0 - 2y - 0}$$

$$\frac{\delta u}{\delta y} = -2y \text{ ---(v)}$$

From (iii)

$$v = 2xy$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(2xy)$$

$$\frac{\delta v}{\delta x} = 2y \text{ --(vi)}$$

Again

$$v = 2xy$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(2xy)$$

$$\frac{\delta v}{\delta y} = 2x \text{ --(vii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \text{ ------(viii)}$$

Putting the values of $\frac{\delta u}{\delta x}, \frac{\delta v}{\delta y}, \frac{\delta u}{\delta y}, \frac{\delta v}{\delta x}$ in (viii)

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x}$$

$$\Rightarrow 2x = 2x \quad \Rightarrow -2y = -2y$$

L.H.S = R.H.S **L.H.S = R.H.S**

Since the Cauchy-Riemann Equations are satisfied by the function $w = f(z) = z^2 - 4$.
Hence the function $w = f(z) = z^2 - 4$ is analytic.

Example 16

Determine the function $w = f(z) = z\bar{z}$ is analytic or not.

Answer:

Given,

$$w = f(z) = z\bar{z}$$

$$u + iv = (x + iy)(x - iy) \quad [\because w = u + iv \ \& \ z = x + iy \ \& \ \bar{z} = x - iy]$$

$$u + iv = x^2 - xiy + xiy - i^2y^2$$

$$u + iv = x^2 + y^2 \quad [i^2 = -1] \text{ --------------------(i)}$$

Equating real and imaginary part,

$$u = x^2 + y^2 \quad \text{---------------------------------------(ii)}$$

$$v = 0 \quad \text{---------------------------------------(iii)}$$

Since the Cauchy-Riemann Equations are satisfied by the function $\mathbf{w = f(z) = e^{x}(\cos y + i \sin y)}$. Hence the function $\mathbf{w = f(z) = e^{x}(\cos y + i \sin y)}$ is analytic.

2nd Part: We have,
From (ii)

$$\mathbf{w = f(z) = u + iv = e^{x}\cos y + ie^{x}\sin y}$$

$$\mathbf{w = f(z) = u + iv} \qquad \text{----------------------------(x)}$$

Differentiating (x) with respect to x

$$\mathbf{f'(z) = \frac{\delta u}{\delta x} + i\frac{\delta v}{\delta x}}$$

$$\mathbf{f'(z) = e^{x}\cos y + ie^{x}\sin y} \qquad \text{[From (v) \& (vii);} \ \frac{\delta u}{\delta x} = e^{x}\cos y, \frac{\delta v}{\delta x} = e^{x}\sin y \text{]}$$

$$\mathbf{f'(z) = e^{x}(\cos y + i\sin y)}$$

$$\mathbf{f'(z) = e^{x}e^{iy}} \qquad \text{[From (i):} \ e^{iy} = \cos y + i\sin y \text{]}$$

$$\mathbf{f'(z) = e^{x+iy}}$$

$$\mathbf{f'(z) = e^{z}} \quad \text{Answer} \qquad [z = x + iy]$$

Example 18

Determine the function $\mathbf{w = f(z) = z^3}$ is analytic or not.

Answer:

Given,

$$\mathbf{w = f(z) = z^3}$$

$$\mathbf{\Rightarrow u + iv = (x + iy)^3} \qquad [\because w = u + iv \ \& \ z = x + iy]$$

$$\mathbf{\Rightarrow u + iv = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3} \qquad [(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3]$$

$$\mathbf{\Rightarrow u + iv = x^3 + i3x^2y - 3xy^2 - iy^3} \qquad [i^2 = -1]$$

$$\mathbf{\Rightarrow u + iv = x^3 - 3xy^2 + i(3x^2y - y^3)} \qquad \text{---------------------------(i)}$$

Equating real and imaginary part

$$\mathbf{u = x^3 - 3xy^2} \qquad \text{-----------------------------(ii)}$$

$$\mathbf{v = 3x^2y - y^3} \qquad \text{-----------------------------(iii)}$$

Differentiating (ii) partially with respect to x

$$\mathbf{u = x^3 - 3xy^2}$$

$$\mathbf{\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x^3 - 3xy^2)}$$

$$\mathbf{\frac{\delta u}{\delta x} = 3x^2 - 3y^2} \qquad \text{-----------------------------(iv)}$$

Again, Differentiating (ii) partially with respect to y

$$\mathbf{u = x^3 - 3xy^2}$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^3 - 3xy^2)$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^3) - 3x\frac{\delta}{\delta y}(y^2)$$

$$\frac{\delta u}{\delta y} = 0 - 6xy$$

$$\frac{\delta u}{\delta y} = -6xy \qquad \text{-----------------------------(v)}$$

Differentiating (iii) partially with respect to x

$$v = 3x^2y - y^3$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(3x^2y - y^3)$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(3x^2y) - \frac{\delta}{\delta x}(y^3)$$

$$\frac{\delta v}{\delta x} = y\frac{\delta}{\delta x}(3x^2) - \frac{\delta}{\delta x}(y^3)$$

$$\frac{\delta v}{\delta x} = 6xy - 0$$

$$\frac{\delta v}{\delta x} = 6xy \qquad \text{-----------------------------(vi)}$$

Again, Differentiating (iii) partially with respect to y

$$v = 3x^2y - y^3$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(3x^2y - y^3)$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(3x^2y) - \frac{\delta}{\delta y}(y^3)$$

$$\frac{\delta v}{\delta y} = 3x^2\frac{\delta}{\delta y}(y) - \frac{\delta}{\delta y}(y^3)$$

$$\frac{\delta v}{\delta y} = 3x^2.1 - 3y^2$$

$$\frac{\delta v}{\delta y} = 3x^2 - 3y^2 \qquad \text{-----------------------------(vii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \qquad \& \qquad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \text{ ----------------(viii)}$$

Putting the values of $\frac{\delta u}{\delta x}, \frac{\delta v}{\delta y}, \frac{\delta u}{\delta y}, \frac{\delta v}{\delta x}$ in (viii)

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \qquad \& \qquad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x}$$

$\Rightarrow 3x^2 - 3y^2 = 3x^2 - 3y^2$ $\qquad\qquad$ $\Rightarrow -6xy = -6xy$

L.H.S = R.H.S $\qquad\qquad$ L.H.S = R.H.S

Since the Cauchy-Riemann Equations are satisfied by the function $\mathbf{w = f(z) = z^3}$. Hence the function $\mathbf{w = f(z) = z^3}$ is analytic.

Home Task:

$\mathbf{f(z) = f(x+iy) = (x^3 - 3xy^2 - 2x) + i(3x^2y - y^3 - 2y)}$ is analytic or not.

Example 19

Derive Laplace's Equation from Cauchy-Riemann Equations

Answer:

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \textit{and} \quad \frac{\delta v}{\delta x} = -\frac{\delta u}{\delta y}$$

That is

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \qquad \text{-----------------------------(i)}$$

$$\frac{\delta v}{\delta x} = -\frac{\delta u}{\delta y} \qquad \text{-----------------------------(ii)}$$

Differentiating (i) with respect to x we get,

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y}$$

$$\frac{\delta}{\delta x}\left(\frac{\delta u}{\delta x}\right) = \frac{\delta}{\delta x}\left(\frac{\delta v}{\delta y}\right)$$

$$\frac{\delta^2 u}{\delta x^2} = \frac{\delta^2 v}{\delta x \delta y} \qquad \text{-----------------------------(iii)}$$

Differentiating (i) with respect to y we get,

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y}$$

$$\frac{\delta}{\delta y}\left(\frac{\delta u}{\delta x}\right) = \frac{\delta}{\delta y}\left(\frac{\delta v}{\delta y}\right)$$

$$\frac{\delta^2 u}{\delta y \delta x} = \frac{\delta^2 v}{\delta y^2}$$

$$\frac{\delta^2 v}{\delta y^2} = \frac{\delta^2 u}{\delta y \delta x} \qquad \text{-----------------------------(iv)}$$

Again

Differentiating (ii) with respect to x we get,

$$\frac{\delta v}{\delta x} = -\frac{\delta u}{\delta y}$$

$$\frac{\delta}{\delta x}\left(\frac{\delta v}{\delta x}\right) = -\frac{\delta}{\delta x}\left(\frac{\delta u}{\delta y}\right)$$

$$\frac{\delta^2 v}{\delta x^2} = -\frac{\delta^2 u}{\delta x \delta y} \qquad \text{-----------------------------(v)}$$

Differentiating (ii) with respect to y we get,

$$\frac{\delta v}{\delta x} = -\frac{\delta u}{\delta y}$$

$$\frac{\delta}{\delta y}\left(\frac{\delta v}{\delta x}\right) = -\frac{\delta}{\delta y}\left(\frac{\delta u}{\delta y}\right)$$

$$\frac{\delta^2 v}{\delta y \delta x} = -\frac{\delta^2 u}{\delta y^2}$$

$$\frac{\delta^2 u}{\delta y^2} = -\frac{\delta^2 v}{\delta y \delta x} \qquad \text{-----------------------------(vi)}$$

Adding (iii) & (vi)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = \frac{\delta^2 \cancel{v}}{\delta \cancel{y} \delta x} - \frac{\delta^2 \cancel{v}}{\cancel{\delta x} \delta y}$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0 \qquad \text{-----------------------------(vii)}$$

Adding (iv) & (v)

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{\delta^2 \cancel{u}}{\delta \cancel{y} \delta x} - \frac{\delta^2 \cancel{u}}{\cancel{\delta x} \delta y}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = 0 \qquad \text{-----------------------------(viii)}$$

The Equation no (vii) & (viii) are called Laplace's Equation

What is Harmonic Function?

Any function which satisfies the Laplace's Equation is known as a harmonic function. If $f(z) = u + jv$ is an analytic function, then u *and* v *are* both *harmonic function..*

A function $\mathbf{f(x, y, z)}$ is called a harmonic function if its second-order partial derivatives exist and if it satisfies Laplace's equation: $\frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} + \frac{\delta^2 f}{\delta z^2} = 0$

Example 20:

Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic functions of (x,y)

Answer: Given

$$\mathbf{u = x^2 - y^2} \qquad \text{-------------------------------(i)}$$

Differentiating (i) with respect to x

$$\frac{\delta \mathbf{u}}{\delta \mathbf{x}} = \frac{\delta}{\delta \mathbf{x}}(\mathbf{x^2 - y^2})$$

$$\frac{\delta \mathbf{u}}{\delta \mathbf{x}} = \mathbf{2x + 0}$$

$$\frac{\delta \mathbf{u}}{\delta \mathbf{x}} = \mathbf{2x} \qquad \text{-------------------------------(ii)}$$

Again differentiating (ii) with respect to x

$$\frac{\delta \mathbf{u}}{\delta \mathbf{x}} = \mathbf{2x}$$

$$\frac{\delta}{\delta \mathbf{x}}\left(\frac{\delta \mathbf{u}}{\delta \mathbf{x}}\right) = \frac{\delta}{\delta \mathbf{x}}(\mathbf{2x})$$

$$\frac{\delta^2 \mathbf{u}}{\delta \mathbf{x}^2} = \mathbf{2} \qquad \text{-------------------------------(iii)}$$

Now differentiating (i) with respect to y

$$\mathbf{u = x^2 - y^2}$$

$$\frac{\delta \mathbf{u}}{\delta \mathbf{y}} = \frac{\delta}{\delta \mathbf{y}}(\mathbf{x^2 - y^2})$$

$$\frac{\delta \mathbf{u}}{\delta \mathbf{y}} = \mathbf{0 - 2y}$$

$$\frac{\delta \mathbf{u}}{\delta \mathbf{y}} = \mathbf{0 - 2y} \qquad \text{-------------------------------(iv)}$$

Again differentiating (iv) with respect to y

$$\frac{\delta \mathbf{u}}{\delta \mathbf{y}} = \mathbf{-2y}$$

$$\frac{\delta}{\delta \mathbf{y}}\left(\frac{\delta \mathbf{u}}{\delta \mathbf{y}}\right) = -\frac{\delta}{\delta \mathbf{y}}(\mathbf{2y})$$

$$\frac{\delta^2 \mathbf{u}}{\delta \mathbf{y}^2} = \mathbf{-2} \qquad \text{-------------------------------(v)}$$

Adding (iii) and (v)

$$\frac{\delta^2 \mathbf{u}}{\delta \mathbf{x}^2} + \frac{\delta^2 \mathbf{u}}{\delta \mathbf{y}^2} = \mathbf{2 - 2 = 0}$$

$$\frac{\delta^2 \mathbf{u}}{\delta \mathbf{x}^2} + \frac{\delta^2 \mathbf{u}}{\delta \mathbf{y}^2} = \mathbf{0}$$

Since $\mathbf{u(x,y) = x^2 - y^2}$ satisfies Laplace's equation. Hence $\mathbf{u(x,y) = x^2 - y^2}$ is a harmonic function.

Now, given

$$v = \frac{y}{x^2 + y^2} \qquad \text{-------------------------------(vi)}$$

Differentiating (vi) with respect to x

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}\left(\frac{y}{x^2 + y^2}\right)$$

$$\frac{\delta v}{\delta x} = \frac{(x^2 + y^2)\frac{\delta}{\delta x}(y) - y\frac{\delta}{\delta x}(x^2 + y^2)}{(x^2 + y^2)^2} \qquad [\because \frac{d}{dx}(\frac{u}{v}) = \frac{v\frac{d}{dx}(u) - u\frac{d}{dx}(v)}{v^2}]$$

$$= \frac{(x^2 + y^2).0 - y(2x + 0)}{(x^2 + y^2)^2}$$

$$= \frac{0 - 2xy}{(x^2 + y^2)^2}$$

$$\frac{\delta v}{\delta x} = \frac{-2xy}{(x^2 + y^2)^2} \qquad \text{-------------------------------(vii)}$$

Again differentiating (vii) with respect to x

$$\frac{\delta}{\delta x}\left(\frac{\delta v}{\delta x}\right) = -\frac{\delta}{\delta x}\{\frac{2xy}{(x^2 + y^2)^2}\}$$

$$\frac{\delta^2 v}{\delta x^2} = -[\frac{(x^2 + y^2)^2\frac{\delta}{\delta x}(2xy) - 2xy\frac{\delta}{\delta x}(x^2 + y^2)^2}{\{(x^2 + y^2)^2\}^2}]$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{-(x^2 + y^2)^2 2y\frac{\delta}{\delta x}(x) + 2xy \times 2(x^2 + y^2)^{2-1}\frac{\delta}{\delta x}(x^2 + y^2)}{(x^2 + y^2)^4}$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{-(x^2 + y^2)^2 2y.1 + 4xy(x^2 + y^2)(2x + 0)}{(x^2 + y^2)^4}$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{-(x^2 + y^2)^2 2y + 8x^2y(x^2 + y^2)}{(x^2 + y^2)^4}$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{(x^2 + y^2)[-(x^2 + y^2)2y + 8x^2y]}{(x^2 + y^2)^3(x^2 + y^2)}$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{[-2x^2y - 2y^3 + 8x^2y]}{(x^2 + y^2)^3}$$

$$\frac{\delta^2 v}{\delta x^2} = \frac{6x^2y - 2y^3}{(x^2 + y^2)^3} \qquad \text{-------------------------------(viii)}$$

Again,

$$v=\frac{y}{x^2+y^2}$$

Differentiating with respect to y

$$\frac{\delta v}{\delta y}=\frac{\delta}{\delta y}\left(\frac{y}{x^2+y^2}\right)$$

$$\frac{\delta v}{\delta y}=\frac{(x^2+y^2)\frac{\delta}{\delta y}(y)-y\frac{\delta}{\delta y}(x^2+y^2)}{(x^2+y^2)^2}$$

$$\frac{\delta v}{\delta y}=\frac{(x^2+y^2).1-y(0+2y)}{(x^2+y^2)^2}$$

$$\frac{\delta v}{\delta y}=\frac{(x^2+y^2)-2y^2}{(x^2+y^2)^2}$$

$$\frac{\delta v}{\delta y}=\frac{(x^2-y^2)}{(x^2+y^2)^2} \qquad \text{-------------------------------(ix)}$$

Again, differentiating (ix) with respect to y

$$\frac{\delta}{\delta y}\left(\frac{\delta v}{\delta y}\right)=\frac{\delta}{\delta y}\left\{\frac{(x^2-y^2)}{(x^2+y^2)^2}\right\}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)^2\frac{\delta}{\delta y}(x^2-y^2)-(x^2-y^2)\frac{\delta}{\delta y}(x^2+y^2)^2}{\{(x^2+y^2)^2\}^2}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)^2(0-2y)-(x^2-y^2)\times 2(x^2+y^2)^{2-1}\frac{\delta}{\delta y}(x^2+y^2)}{(x^2+y^2)^4}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)^2(0-2y)-(x^2-y^2)\times 2(x^2+y^2)(0+2y)}{(x^2+y^2)^4}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)^2(-2y)-2(x^2-y^2)(x^2+y^2)(2y)}{(x^2+y^2)^4}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)^2(-2y)-4y(x^2-y^2)(x^2+y^2)}{(x^2+y^2)^4}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)[(x^2+y^2)(-2y)-4y(x^2-y^2)]}{(x^2+y^2)(x^2+y^2)^3}$$

$$\frac{\delta^2 v}{\delta y^2}=\frac{(x^2+y^2)(-2y)-4y(x^2-y^2)}{(x^2+y^2)^3}$$

$$\frac{\delta^2 v}{\delta y^2} = \frac{-2x^2y - 2y^3 - 4x^2y + 4y^3}{(x^2+y^2)^3}$$

$$\frac{\delta^2 v}{\delta y^2} = \frac{-6x^2y + 2y^3}{(x^2+y^2)^3} \qquad \text{-------------------------------(x)}$$

Adding (viii) and(x),

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{6x^2y - 2y^3}{(x^2+y^2)^3} + \frac{-6x^2y + 2y^3}{(x^2+y^2)^3}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{6x^2y - 2y^3 - 6x^2y + 2y^3}{(x^2+y^2)^3}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = 0$$

Since $v(x,y) = \frac{y}{x^2+y^2}$ satisfies Laplace's equation. Hence $v(x,y) = \frac{y}{x^2+y^2}$ is a harmonic function. ***(Proved)***

Example 21:

Prove that, $u = e^{-x}(x\sin y - y\cos y)$ is a harmonic function.

Answer:

Given, $u = e^{-x}(x\sin y - y\cos y)$.....................................(i)

Differentiating (i) with respect to x,

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}[e^{-x}(x\sin y - y\cos y)]$$

$$\frac{\delta u}{\delta x} = e^{-x}\frac{\delta}{\delta x}(x\sin y - y\cos y) + (x\sin y - y\cos y)\frac{\delta}{\delta x}(e^{-x}) \; [\frac{d}{dx}(uv) = u\frac{d}{dx}v + v\frac{d}{dx}u]$$

$$\frac{\delta u}{\delta x} = e^{-x}[\frac{\delta}{\delta x}(x\sin y) - \frac{\delta}{\delta x}(y\cos y)] - (x\sin y - y\cos y)e^{-x} \;\; [\because \frac{d}{dx}(e^{-x}) = -e^{-x}]$$

$$\frac{\delta u}{\delta x} = e^{-x}[\{x\frac{\delta}{\delta x}\sin y + \sin y\frac{\delta}{\delta x}x\} - \{y\frac{\delta}{\delta x}\cos y + \cos y\frac{\delta}{\delta x}y\}] - e^{-x}(x\sin y - y\cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x}[\{x(0) + \sin y.1\} - \{y(0) + \cos y.(0)\}] - e^{-x}(x\sin y - y\cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x}[0 + \sin y - 0 - 0] - e^{-x}(x\sin y - y\cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x}\sin y - e^{-x}(x\sin y - y\cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x}\sin y - xe^{-x}\sin y + ye^{-x}\cos y$$

$$\frac{\delta u}{\delta x} = e^{-x}(\sin y - x\sin y + y\cos y) \text{.....................................(ii)}$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x}[\frac{\delta}{\delta y}(x\cos y) + \frac{\delta}{\delta y}(y\sin y) - \frac{\delta}{\delta y}(\cos y)] + (x\cos y + y\sin y - \cos y)\frac{\delta}{\delta y}(e^{-x})$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x}[x\frac{\delta}{\delta y}(\cos y) + \cos y\frac{\delta}{\delta y}x + y\frac{\delta}{\delta y}(\sin y) + \sin y\frac{\delta}{\delta y}y - \frac{\delta}{\delta y}(\cos y)] + (x\cos y + y\sin y - \cos y).0$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x}[-x\sin y + \cos y.0 + y\cos y + \sin y + \sin y] + 0$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x}[-x\sin y + y\cos y + 2\sin y]\text{...}(v)$$

Adding (iii) and (v),

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^{-x}[-2\sin y + x\sin y - y\cos y] + e^{-x}[2\sin y - x\sin y + y\cos y]$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^{-x}[-2\sin y + x\sin y - y\cos y + 2\sin y - x\sin y + y\cos y]$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$$

Since $u(x,y) = e^{-x}(x\sin y - y\cos y)$ satisfies Laplace's equation,

hence $u(x,y) = e^{-x}(x\sin y - y\cos y)$ is a Harmonic function.

Home Task

i. Prove that $f(z) = |z|^2$ is not harmonic functions but $f(z) = \ln(|z|^2)$ is harmonic.
ii. **Is** $f(x,y,z) = x^2 + y^2 - 2z^2$ harmonic? What about $f(x,y,z) = x^2 - y^2 + z^2$?
iii. **Show that the function** $u(x,y) = 3x^3 - 9xy^2$ **is harmonic**
iv. **Verify** $u(x,y) = x^3 - 3xy^2 - 5y$ **is harmonic**
v. **Test the following functions harmonic or not**
   a) $u = x^3 - 3xy^2$
   b) $u = e^{-y}\sin x$
   c) $u = e^x\cos y$
   d) $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$
   e) $u(x,y) = e^x(x\cos y - y\sin y)$
   f) $u = y^3 - 3x^2y$
   g) $u = 2x(1-y)$
   h) $u(x,y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$
   i) $u(x,y) = x^2 - y^2 - 2xy - 2x + 3y$

$dz = dx$ ....(iii)

$\int_C z\,dz$

$= \int_{AC} (x + iy)\,dx$

$= \int_{AC} (x + i.1)\,dx$

$= \int_1^3 (x + i.1)dx$

$= [\frac{x^2}{2} + ix]_1^3$

$= [\frac{3^2}{2} + 3i - \frac{1}{2} - i]$

$= [\frac{9}{2} + 3i - \frac{1}{2} - i]$

$= [\frac{8}{2} + 2i]$

$= [4 + 2i]$

What is pole: pole is a certain type of singularity of a function can be found by substituting the denominator of the function equal to zero. Roots of denominator indicates poles. That is, Poles represents the points where a complex function cease to be analytic.

Cauchy's Theorem

The theorem states that if $f(z)$ is analytic everywhere within a simply-connected region then $\oint_c f(z)dz = 0$ for every simple closed path C lying in the region.

Cauchy's Integral Formula:

If $f(z)$ is analytic inside and on the boundary C of a simply-connected region then for any point 'a' within the curve 'C': $\oint_c \frac{f(z)}{z - a} dz = 2\pi i \times f(a)$

Example 27:

Evaluate $\int_c \frac{z^2 - z + 1}{z - 1} dz$

Where c is the circle i) $|z| = 1$ ii) $|z| = \frac{1}{2}$

We have,

$z = x + jy$

$\therefore |z| = \sqrt{x^2 + y^2}$

Given,

i) $|z| = 1$

$\sqrt{x^2 + y^2} = 1$

$\therefore x^2 + y^2 = 1$

$(x-0)^2 + (y-0)^2 = 1^2$ ----------------(i)

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

Which is the equation of a circle whose Center (0, 0), Radius =1

Let $f(z) = z^2 - z + 1$ and singularity point is $a = 1$

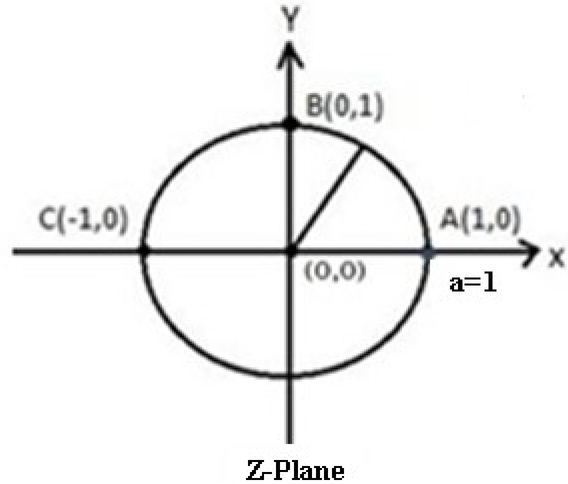


Figure 47

Pole $z - 1 = 0$

$\Rightarrow z = 1$

$\Rightarrow x + jy = 1 + 0.j$

Equating the coefficient of real and imaginary part, we get,

$$x = 1, \qquad y = 0$$

Hence, we can write,

$$(x, y) = (1,0)$$

$\because$ $(x, y) = (1,0)$ is on the circle c, then by Cauchy's Integral formula

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

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$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times f(1) \; [a = 1]$$

Here, $f(z) = z^2 - z + 1$

$\therefore f(1) = 1^2 - 1 + 1 = 1$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times f(1)$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times 1$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \; \textit{Answer}$$

Given,

i) $|z| = \frac{1}{2}$

$\sqrt{x^2 + y^2} = \frac{1}{2}$

$\therefore x^2 + y^2 = \frac{1}{4}$

$(x - 0)^2 + (y - 0)^2 = (\frac{1}{2})^2$ ----------------(i)

[We have, $(x - a)^2 + (y - b)^2 = r^2$]

Which is the equation of a circle whose Center (0, 0), Radius = $\frac{1}{2}$

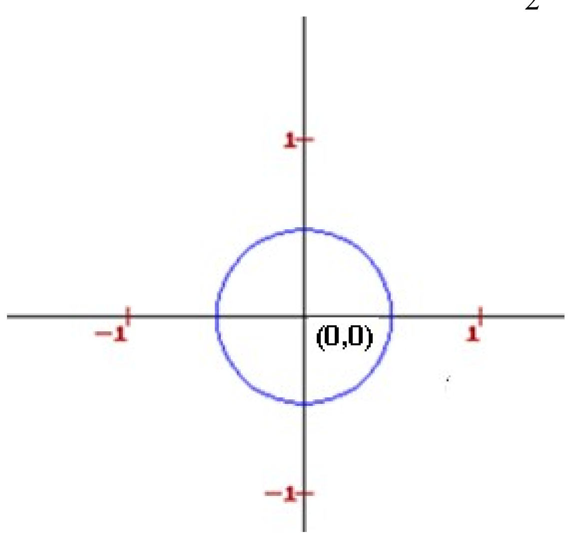


Figure 48

Pole $z - 1 = 0$

$\Rightarrow \mathbf{z = 1}$

$\Rightarrow \mathbf{x + jy = 1 + 0.j}$

Equating the coefficient of real and imaginary part, we get,

$$\mathbf{x = 1, \qquad y = 0}$$

Hence, we can write,

$$\mathbf{(x, y) = (1,0)}$$

$\because$ $\mathbf{(x, y) = (1,0)}$ is outside the circle c, then by Cauchy's theorem

$$\int_c \mathbf{f(z)dz = 0}$$

$$\therefore \int_c \frac{\mathbf{z^2 - z + 1}}{\mathbf{z - 1}} \mathbf{dz = 0}$$

Example 28:

Evaluate $\int_c \frac{z}{z^2 - 3z + 2} dz$

Where c is the circle $|\mathbf{z - 2}| = \frac{\mathbf{1}}{\mathbf{2}}$

We have,

$\mathbf{z = x + jy}$

$\mathbf{z - 2 = x + jy - 2}$

$\mathbf{z - 2 = x - 2 + jy}$

$\therefore |\mathbf{z - 2}| = \sqrt{\mathbf{(x - 2)^2 + y^2}}$

Given,

$|\mathbf{z - 2}| = \frac{\mathbf{1}}{\mathbf{2}}$

$\therefore |\mathbf{z - 2}| = \sqrt{\mathbf{(x - 2)^2 + y^2}} = \frac{\mathbf{1}}{\mathbf{2}}$

$\therefore \mathbf{(x - 2)^2 + y^2} = \frac{1}{4}$

$\therefore \mathbf{(x - 2)^2 + (y - 0)^2 = (\frac{1}{2})^2}$

Which is the equation of a circle whose Center (2, 0), Radius $= \frac{\mathbf{1}}{\mathbf{2}}$

$$\int_c \frac{f(z)}{z}dz = \int_c \frac{\frac{2z+1}{z+1}}{z}dz = 2\pi i$$

Example 30.Show that $\oint_c \frac{e^{tz}}{z^2+1}dz = 2\pi i \sin t$ **, where c is the circle** $|z| = 3$

Answer: Now, Poles: $z^2 + 1 = 0$

$\Rightarrow z^2 = -1$

$\Rightarrow z^2 = i^2$

$\Rightarrow z = \pm i$

$\Rightarrow x + jy = 0 \pm 1.j$

Equating the coefficient of real and imaginary part, we get,

$x = 0, \quad y = 1 \quad \text{or} \quad x = 0, \quad y = -1$

Hence, we can write,

$$(x, y) = (0,1) \quad \text{or} \quad (x, y) = (0,-1)$$

Given

$|z| = 3$

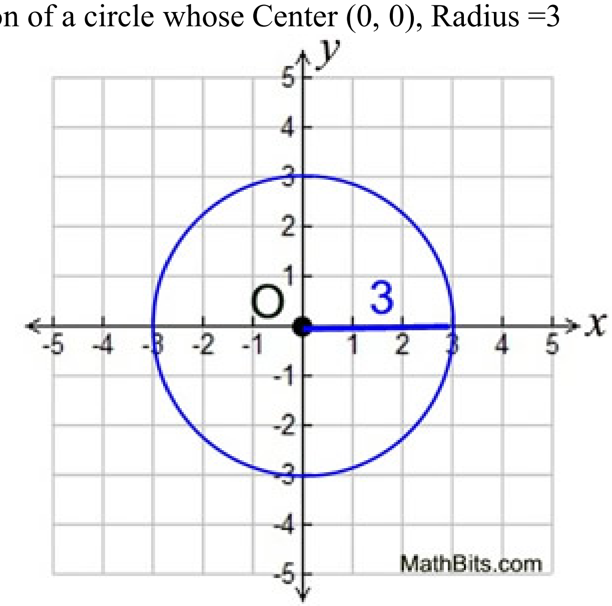
$\Rightarrow \sqrt{x^2 + y^2} = 3$

$\Rightarrow x^2 + y^2 = 9$

$\Rightarrow x^2 + y^2 = 3^2$

$\Rightarrow x^2 + y^2 = 3^2$

Which is the equation of a circle whose Center (0, 0), Radius =3



$\because$ $(x,y)=(0,1)$ and **or** $(x,y)=(0,-1)$ is on the circle c, then by Cauchy's Integral formula

$$\int_c \frac{f(z)}{z-a}dz = 2\pi i \times f(a)$$

$$\int_c \frac{f(z)}{z-i}dz = 2\pi i \times f(i) ----------------(i)$$

And

$$\int_c \frac{f(z)}{z+i}dz = 2\pi i \times f(-i) ----------------(ii)$$

Given,

$$\oint_c \frac{e^{tz}}{z^2+1}dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)}dz = \oint_c e^{tz}\frac{1}{(z+i)(z-i)}dz$$

$$\oint_c \frac{e^{tz}}{z^2+1}dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)}dz = \frac{1}{2i}\oint_c e^{tz}[\frac{1}{(z-i)} - \frac{1}{(z+i)}]dz$$

$$\oint_c \frac{e^{tz}}{z^2+1}dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)}dz = \frac{1}{2i}\oint_c [\frac{e^{tz}}{(z-i)} - \frac{e^{tz}}{(z+i)}]dz$$

$$\oint_c \frac{e^{tz}}{z^2+1}dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)}dz = \frac{1}{2i}[\oint_c \frac{e^{tz}}{(z-i)}dz - \oint_c \frac{e^{tz}}{(z+i)}dz]$$

We have, Cauchy's Integral Formula:

$$\int_c \frac{f(z)}{z-a}dz = 2\pi i \times f(a)$$

For $\oint_c \frac{e^{tz}}{(z-i)}dz$

$f(z) = e^{tz}$

$\therefore f(i) = e^{it} ----------------(iii)$

And

For $\oint_c \frac{e^{tz}}{(z+i)}dz$

$f(z) = e^{tz}$

$\therefore f(-i) = e^{-it} ----------------(iv)$

$$\oint_c \frac{e^{tz}}{z^2+1}dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)}dz = \frac{1}{2i}[\oint_c \frac{e^{tz}}{(z-i)}dz - \oint_c \frac{e^{tz}}{(z+i)}dz]$$

$$= \frac{1}{2i}[2\pi i \times f(i) - 2\pi i \times f(-i)] \quad [\because \int_c \frac{f(z)}{z-a}dz = 2\pi i \times f(a)]$$

$$= \frac{1}{2i}[2\pi i \times e^{it} - 2\pi i \times e^{-it}]$$

$$= 2\pi i \frac{1}{2i}[e^{it} - e^{-it}]$$

$$= 2\pi i \times \sin t \quad \text{(Proved)} \qquad [\because \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})]$$

We have,

$$e^{ix} = \cos x + i \sin x$$

$$e^{-ix} = \cos x - i \sin x$$

$$e^{ix} - e^{-ix} = 2i \sin x$$

$$\therefore \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

Example 31: Show that $\oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$ **, where c is the circle** $|z| = 5$

Poles: $z + \frac{\pi}{2} = 0$

$$\Rightarrow z = -\frac{\pi}{2}$$

Poles: $z + \frac{\pi}{2} = 0$

$$\Rightarrow z = -\frac{\pi}{2}$$

$$\Rightarrow x + jy = -\frac{\pi}{2} + j.0$$

So, $(x, y) = (-\frac{\pi}{2}, 0)$

Given

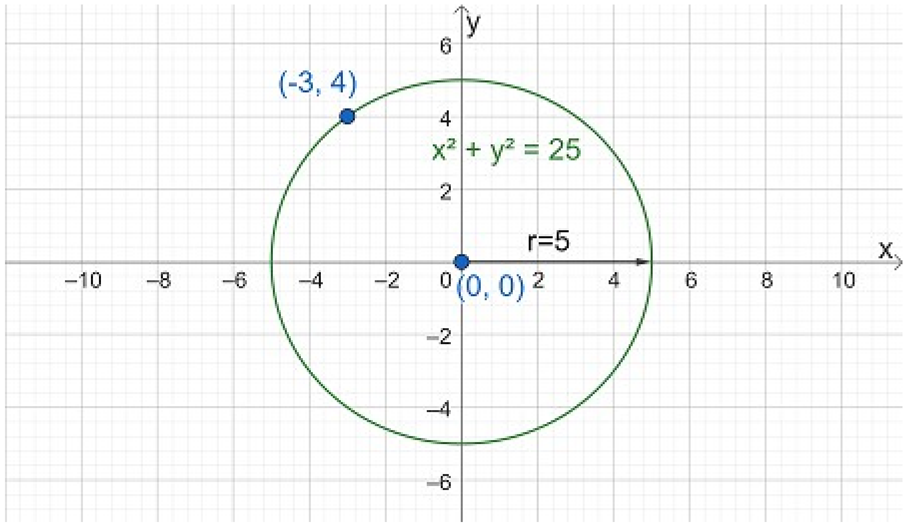
$$|z| = 5$$

$$\Rightarrow \sqrt{x^2 + y^2} = 5$$

$$\Rightarrow x^2 + y^2 = 25$$

$$\Rightarrow x^2 + y^2 = 5^2$$

Which is the equation of a circle whose Center (0, 0), Radius =5



$\because$ $(x, y) = (-\frac{\pi}{2}, 0)$ is on the circle c, then by Cauchy's Integral

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

Here,

$$\oint_c \frac{\sin 3z}{z+\frac{\pi}{2}} dz$$

$$f(z) = \sin 3z$$

$$\therefore f(-\frac{\pi}{2}) = \sin(-3\frac{\pi}{2})$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3\frac{\pi}{2}) \qquad [\sin(-\theta) = -\sin\theta]$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3 \times \frac{\pi}{2} + 0)$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3 \times 90^0 + 0)$$

$$\therefore f(-\frac{\pi}{2}) = -(-\cos 0)$$

$$\therefore f(-\frac{\pi}{2}) = 1$$

$$\therefore \oint_c \frac{\sin 3z}{z+\frac{\pi}{2}} dz = 2\pi i \times f(-\frac{\pi}{2})$$

$$\therefore \oint_c \frac{\sin 3z}{z+\frac{\pi}{2}} dz = 2\pi i \times 1$$

$$\therefore \oint_c \frac{\sin 3z}{z+\frac{\pi}{2}}dz = 2\pi i \text{ (Proved)}$$

Example 32:

Find the residue at pole of $\dfrac{1-2z}{z(z-1)(z-2)}$

Answer:

Let, $f(z) = \dfrac{1-2z}{z(z-1)(z-2)}$

Poles $z(z-1)(z-2) = 0$

$z = 0;\ (z-1) = 0;\ (z-2) = 0$

$z = 0;\ z = 1;\ z = 2$

Residue of $f(z)$ at $(z = 0)$ $= \lim\limits_{z\to 0}(z-0)f(z)$

$$= \lim_{z\to 0} z\frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z\to 0} \frac{1-2z}{(z-1)(z-2)}$$

$$= \frac{1}{(-1)(-2)}$$

$$= \frac{1}{2}$$

Residue of $f(z)$ at $(z = 1)$ $= \lim\limits_{z\to 1}(z-1)f(z)$

$$= \lim_{z\to 1} (z-1)\frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z\to 1} \frac{1-2z}{z(z-2)}$$

$$= \frac{1-2}{(1)(1-2)}$$

$$= 1$$

Residue of $f(z)$ at $(z = 2)$ $= \lim\limits_{z\to 2}(z-2)f(z)$

$$= \lim_{z\to 2} (z-2)\frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z\to 2} \frac{1-2z}{z(z-1)}$$

$$= \frac{1-2\times 2}{(2)(2-1)}$$

$$= \frac{1-2\times 2}{(2)(2-1)}$$

$$= -\frac{3}{2}$$

Example 33: Consider the function $f(z)=\frac{5}{i-3z}$; to find the poles and residue of the function

Answer:

Given $f(z)=\frac{5}{i-3z}$

To find the pole and residue , divide the numerator and denominator by -3

$$\therefore f(z)=\frac{-\frac{5}{3}}{z-\frac{i}{3}}$$

That is the pole of the function $z=\frac{i}{3}$

And residue of $f(z)$ at $(z=\frac{i}{3})$ $=\lim\limits_{z\to\frac{i}{3}}(z-\frac{i}{3})f(z)=\lim\limits_{z\to\frac{i}{3}}(z-\frac{i}{3})\ \frac{5}{i-3z}$

$$=\lim_{z\to\frac{i}{3}}(\frac{3z-i}{3})\ \frac{5}{i-3z}$$

$$=\lim_{z\to\frac{i}{3}}(\frac{3z-i}{3})\ \frac{5}{-(3z-i)}$$

$$=-\frac{5}{3}$$

Or

Given $f(z)=\frac{5}{i-3z}$

Pole : $i-3z=0$

Pole : $-3z=-i$

Pole : $3z=i$

Pole : $z=i/3$

Given $f(z)=\frac{5}{i-3z}$

To find the residue , divide the numerator and denominator by -3

$$\therefore f(z) = \frac{-\frac{5}{3}}{z - \frac{i}{3}}$$

Hence, residue of $f(z) = -\frac{5}{3}$

Example 34: Consider the function $f(z) = \frac{z}{z^2+1}$; to find the poles and residue of the function

Given, $f(z) = \frac{z}{z^2+1}$

To find the pole and residue , we factorize the denominator $f(z) = \frac{z}{(z+i)(z-i)}$

Poles : $z^2 + 1 = 0$

Poles $z^2 = -1$

Poles: $z^2 = i^2$

Poles: $z = \pm i$

Hence there are two simple poles at $z = +i, -i$

And residue of $f(z)$ at $(z = i)$ $= \lim_{z \to i} (z-i) f(z) = \lim_{z \to i} (z-i) \frac{z}{(z+i)(z-i)}$

$$= \lim_{z \to i} \frac{z}{(z+i)}$$

$$= \lim_{z \to i} \frac{i}{(i+i)} = \frac{i}{2i} = \frac{1}{2}$$

Or

To find the residue, at $z = i$, we separate the divergent part to obtain $f(z) = \frac{z}{(z+i)(z-i)}$

We can write, $f(z) = \frac{\frac{z}{z+i}}{(z-i)}$

Hence residue : $\mathrm{Res}[f(z)]_{z=i} = [\frac{z}{(z+i)}]_{z=i} = \frac{i}{i+i} = \frac{1}{2}$

And

To find the residue, at $z = -i$, we separate the divergent part to obtain

$$f(z) = \frac{z}{(z+i)(z-i)}$$

$$\mathbf{f(z) = \frac{\frac{z}{z-i}}{(z+i)}}$$

Hence residue : Res $\mathbf{[f(z)]_{z=-i} = [\frac{z}{(z-i)}]_{z=-i} = \frac{-i}{-i-i} = \frac{1}{2}}$

https://www1.spms.ntu.edu.sg/~ydchong/teaching/09_contour_integration.pdf
file:///C:/Users/REZAUL/Downloads/basic_complex_int_HELM%20(1).pdf

https://complex-analysis.com/content/complex_integration.html
https://www.academia.edu/39133549/Complex_Variables_with_Applications?email_work_card=title (vvvvvvvvvvvvvvv imp)

That is, $s \in A \bigcap B = \underbrace{\{3,4\}}_{s}$

$s \in A$ and $s \in B$

Q-7: Prove De-Morgan's theorem:

1) $(\mathbf{A} \bigcup \mathbf{B})' = \mathbf{A}' \bigcap \mathbf{B}'$

2) $(\mathbf{A} \bigcap \mathbf{B})' = \mathbf{A}' \bigcup \mathbf{B}'$

A= {1,2,3}
B={1,2,3}
B SUBSET OF A
A IS SUBSET OF B
A=B

Proof (1):

Let $\mathbf{x} \in (\mathbf{A} \cup \mathbf{B})'$

$\Rightarrow \mathbf{x} \notin (\mathbf{A} \cup \mathbf{B})$

$\Rightarrow \mathbf{x} \notin \mathbf{A} \quad \textbf{and} \quad \mathbf{x} \notin \mathbf{B}$

$\Rightarrow \mathbf{x} \in \mathbf{A}' \quad \textbf{and} \quad \mathbf{x} \in \mathbf{B}'$

$\Rightarrow \mathbf{x} \in (\mathbf{A}' \cap \mathbf{B}')$

By definition of subset, we get $(\mathbf{A} \bigcup \mathbf{B})' \subseteq \mathbf{A}' \bigcap \mathbf{B}'$----------------------------(1)

Again Let $\mathbf{y} \in \mathbf{A}' \cap \mathbf{B}'$

Then $\mathbf{y} \in \mathbf{A}' \quad \textbf{and} \quad \mathbf{y} \in \mathbf{B}'$

$\Rightarrow \mathbf{y} \notin \mathbf{A} \quad \textbf{and} \quad \mathbf{y} \notin \mathbf{B}$

$\Rightarrow \mathbf{y} \notin (\mathbf{A} \cup \mathbf{B})$

$\Rightarrow \mathbf{y} \in (\mathbf{A} \cup \mathbf{B})'$

By definition of subset, we get $\mathbf{A}' \bigcap \mathbf{B}' \subseteq (\mathbf{A} \bigcup \mathbf{B})'$-----------------------------(2)

From (1) and (2), we get,
$(\mathbf{A} \bigcup \mathbf{B})' = \mathbf{A}' \bigcap \mathbf{B}'$ (Proved)

Proof (2):

Let $\mathbf{x} \in (\mathbf{A} \cap \mathbf{B})'$

$\Rightarrow \mathbf{x} \notin (\mathbf{A} \cap \mathbf{B})$

$\Rightarrow \mathbf{x} \notin \mathbf{A} \quad \textbf{and} \quad \mathbf{x} \notin \mathbf{B}$

$\Rightarrow \mathbf{x} \in \mathbf{A}' \quad \textbf{and} \quad \mathbf{x} \in \mathbf{B}'$

$\Rightarrow \mathbf{x} \in (\mathbf{A}' \cup \mathbf{B}')$

By definition of subset, we get $(\mathbf{A} \bigcap \mathbf{B})' \subseteq \mathbf{A}' \bigcup \mathbf{B}'$ ------------------------------(1)

Again Let $y \in A' \cup B'$

Then $\mathbf{y} \in \mathbf{A}'$ **and** $\mathbf{y} \in \mathbf{B}'$

$\Rightarrow \mathbf{y} \notin \mathbf{A}$ **and** $\mathbf{y} \notin \mathbf{B}$

$\Rightarrow \mathbf{y} \notin (\mathbf{A} \cap \mathbf{B})$

$\Rightarrow \mathbf{y} \in (\mathbf{A} \cap \mathbf{B})'$

By definition of subset, we get $\mathbf{A}' \bigcup \mathbf{B}' \subseteq (\mathbf{A} \bigcap \mathbf{B})'$-----------------------------(2)

From (1) and (2), we get,

$(\mathbf{A} \bigcap \mathbf{B})' = \mathbf{A}' \bigcup \mathbf{B}'$ (Proved)

Q-8: If $A = \{0, \{0,1\}\}$, Find the power set of A.

Answer: Here A contains two elements 0 and the set $\{0,1\}$, Therefore $\mathbf{2^A}$ contain $\mathbf{2^2} = \mathbf{4}$ elements.

i.e. $\mathbf{2^A} = \{\mathbf{A}, \{\mathbf{0}\}, \{\mathbf{0,1}\}, \phi\}$

Q-9: If $E = \{0,1\}$ whether the following statements are true or false

i) $\phi \in \mathbf{E}$ **ii)** $\phi \subset \mathbf{E}$ **iii)** $\phi \in \mathbf{2^E}$ **iv)** $\phi \subset \mathbf{2^E}$ **v)** $\{\phi\} \subset \mathbf{2^E}$

Answer:

Here, $\mathbf{E} = \{\mathbf{0,1}\}$

and $\mathbf{2^E} = \{\mathbf{E}, \{\mathbf{0}\}, \{\mathbf{1}\}, \phi\}$

So,

i) $\phi \in \mathbf{E} \rightarrow$ **False** **ii)** $\phi \subset \mathbf{E} \rightarrow$ **True** **iii)** $\phi \in \mathbf{2^E} \rightarrow$ **True** **iv)** $\phi \subset \mathbf{2^E} \rightarrow$ **False**

v) $\{\phi\} \subset \mathbf{2^E} \rightarrow$ **True**

Q-10: Let P (A) denote set of $\mathbf{A} = \{\mathbf{1,2}\}$, find P {P (A)}.

Solution: Here A contains two elements 1 and 2, therefore $\mathbf{2^A}$ contains $\mathbf{2^2} = \mathbf{4}$ elements, i.e. P (A) or $\mathbf{2^A}$ = {A, {1}, {2}, ϕ}

Again, P {P (A)} contains $\mathbf{2^4} = \mathbf{16}$ elements.

P {P (A)} = {P(A), {A,{1}}, {A,{2}}, {A, ϕ}, {{1},{2}}, {{1}, ϕ}, {{2}, ϕ}, {A,{1},{2}}, {A,{1}, ϕ}, {A,{2}, ϕ}, {{1},{2}, ϕ}, {A}, {{1}}, {{2}}, { ϕ}, ϕ}

Q-11: If A = {1, 2, 3}, B = { $x \mid x$ is even integer} and C = { $x \mid x$ is divisible by 3} are three sets, Find the value of $\mathbf{A} \cup (\mathbf{B} \cap \mathbf{C})$ **and** $\mathbf{A} \cap (\mathbf{B} \cup \mathbf{C})$

Answer:

Here, A = {1, 2, 3},

B = {-6, -4, -2, --------------, 2, 4, 6, 8, ----------∞}

and C = {-6, -3, 3, 6, 12, -------------------------------∞}

Now, $\mathbf{B \cap C} = \{-12,-6,6,12,----------\}$

$\mathbf{A \cup (B \cap C)}$ = {1, 2, 3, -6, -12, 6, 12, 18, ----------------------}

$\mathbf{A \cup (B \cap C)}$ = { -6, -12,1, 2, 3, 6, 12, 18, ----------------------}

Again, $\mathbf{B \cup C}$ = {-6, -4, -3, -2, ----------, 2, 3, 4, 6, 8, 9, 10, 12, ------------}

$\mathbf{A \cap (B \cup C) = \{2,3\}}$ Answer

Q-12: If A, B, C are the sets of integer and
A = $\{\mathbf{x | x^2 - 7x + 12 = 0\}, B = \{x | 2 < x < 12\}, C = \{x | x}$ is divisible by 2} then find the values of $\mathbf{i) A \cap (B \cup C)}$ $\mathbf{ii) (A \cup B) \cap}$ C

Answer:

Here, A = $\mathbf{\{x | x^2 - 7x + 12 = 0\} = \{3,4\}}$

$\mathbf{B = \{x | 2 < x < 12\} = \{3,4,5,6,7,8,9,10,11\}}$

C = {-6, -4, -2, --------------, 2, 4, 6, 8, ----------∞}

We have, $\mathbf{(B \cup C)}$ = {-6, -4, -2, ----------, 2,3,4,5,6,7,8,9,10,11,12,14, ------------
------}

$\mathbf{i) A \cap (B \cup C) = \{3,4\} = A}$

Again $\mathbf{(A \cup B) = (3,4,5,6,7,8,9,10,11\} = B}$

Hence, $\mathbf{(A \cup B) \cap}$ C = {4, 6, 8, 10} Answer

Q-13: Prove that (B-A) is a subset of A'

Answer: Let $\mathbf{x \in B - A}$

$\mathbf{\Rightarrow x \in B}$ **and** $\mathbf{x \notin A}$

$\mathbf{\Rightarrow x \in B}$ **and** $\mathbf{x \in A'}$

Since $\mathbf{x \in B - A}$ **implies** $\mathbf{x \in A'}$

Therefore B-A is a subset of A'

$\mathbf{\therefore B - A \subset A'}$ (Proved)

Q-14: Prove that $\mathbf{B - A' = B \cap A}$

Answer:

Let $\mathbf{x \in B - A'}$

Then,

$\Rightarrow \mathbf{x \in B \text{ and } x \notin A'}$

$\Rightarrow \mathbf{x \in B \text{ and } x \in A}$

$\Rightarrow \mathbf{x \in (B \cap A)}$

$\therefore \mathbf{(B - A') \subseteq (B \cap A)} ---------------- \mathbf{(1)}$

Conversely, Let $\mathbf{y \in B \cap A}$

$\Rightarrow y \in B \text{ and } y \in A$

$\Rightarrow \mathbf{y \in B \text{ and } y \notin A'}$

$\Rightarrow \mathbf{y \in B - A'}$

$\therefore \mathbf{(B \cap A) \subseteq B - A'} ----------------- \mathbf{(2)}$

From (1) and (2), we get,

$\mathbf{B - A' = B \cap A}$ (Proved)

Functions

Definition: Suppose that to each element in a set A there is assigned, by some manner or other, at least a unique element of a set B. We call such assignments a function. Let f denote these assignments, and then we write $\mathbf{f : A \to B}$ (reads f is a function of A into B). Every element of domain should have corresponding image. Each element of domain cannot have different image. The set A is called the domain of the function f, and B is called the co-domain of f. Further if $a \in A$ then the element in B which is assigned to a is called the image of a and is denoted by f (a) (reads f of a)

One input cannot have two corresponding outputs

1. Let f assign to each real number its square, that is, for every real number x let $\mathbf{f(x) = x^2}$. The domain and co-domain of f are both the real numbers, so we can write: $\mathbf{f : R^{\#} \to R^{\#}}$
   The image of -3 is 9; hence we can also write $\mathbf{f(-3) = 9 \text{ or } f : -3 \to 9}$
2. Let f assign to each country in the world its capital city. Here the domain of f is the set of countries in the world; the co-domain of f is the list of capital cities in the world. The image of France is Paris, that is f (France) = Paris.

Let A and B be two sets. A function f from A to B is a relation between A and B such that for each a ∈ A there is one and only one associated b ∈ B. The set A is called the domain of the function; B is called its range.

d

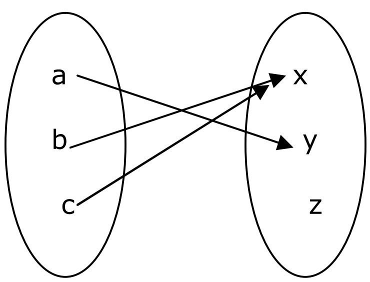
Note that f (A) = {x , y , z } = B
that is, the range of f is equal to the co-domain B. Thus f maps A onto B, i.e f is an onto mapping.

Constant Functions:
A function f of A into B that is, let $\mathbf{f : A \to B}$ is called a constant function if the same element $b \in B$ is assigned to every element in A. i.e. if the range of f consists of only one element.

Example:
Let the function $\mathbf{f : R^{\#} \to R^{\#}}$ be defined by the formula $\mathbf{f(x) = 5}$. Then f is a constant function since 5 is assigned to every element.

Inverse of a function:
Let f be a function of A into B and let $b \in B$. Then the inverse of b, denoted by $\mathbf{f^{-1}(b)}$ consists of those elements in A which are mapped onto b, that is, those elements in A which have b as their image. More briefly, if $\mathbf{f : A \to B}$ then
$\mathbf{f^{-1}(b) = \{x | x \in A, f(x) = b\}}$

Example:
1.



Then $f^{-1}(x)$ = {b, c}, since both b and c have x as their image point. Also $f^{-1}(y)$ = {a}, as only a is mapped into y. The inverse of z, $f^{-1}(z)$, is the null set ϕ, since no element of A is mapped into z.

1. Let the function $\mathbf{f : R^{\#} \to R^{\#}}$ be defined by the formula $\mathbf{f(x) = x^2}$. Then $\mathbf{f^{-1}(4)}$ = {2, -2}, since 4 is the image of both 2 and -2 and there is no other real number whose square is four. Notice that $\mathbf{f^{-1}(-3)}$ $= \phi$, since there is no element in $R^{\#}$ whose square is -3.

Inverse Function:

If $\mathbf{f : A \to B}$ is a one-one function and an onto function, then $\mathbf{f^{-1} : B \to A}$

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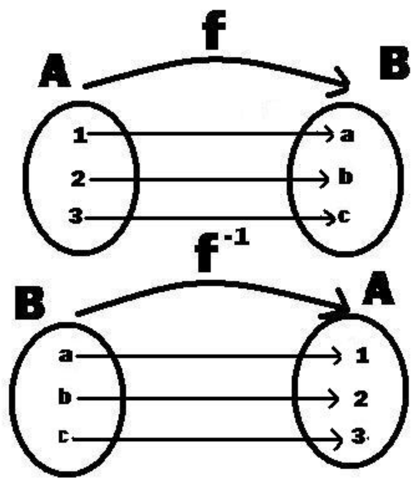


Note that f is one-one and onto. Therefore f^{-1}, the inverse function, exists

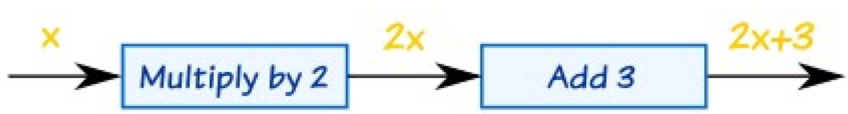
Definition: The inverse of a function is the set of ordered pairs obtained by interchanging the first and second elements of each pair in the original function. If f is a given function, then f^{-1} denotes the inverse of f. (If the original function is a one-to-one function, the inverse will also be a function.)

Inverse Functions

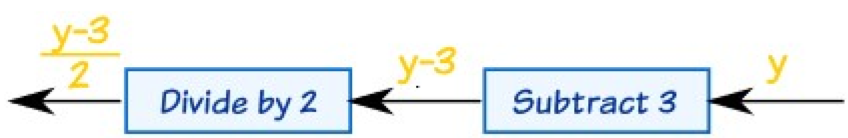
An inverse function goes the other way!

Let us start with an example:

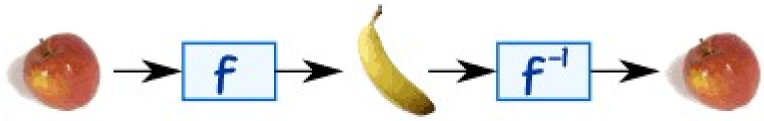
Here we have the function **f(x) = 2x+3**, written as a flow diagram:



The **Inverse Function** just goes the other way:



So the inverse of: 2x+3 is: (y-3)/2



When the function f turns the apple into a banana,
Then the **inverse** function f^{-1} turns the banana back to the apple

Examples:

a. Given function f, find the inverse. Is the inverse also a function?

$f(x) = \{(3,4),(1,-2),(5,-1),(0,2)\}$

Answer:
Function f is a one-to-one function since the x and y values are used only once. The inverse is

$f^{-1}(x) = \{(4,3),(-2,1),(-1,5),(2,0)\}$

Since function f is a one-to-one function, the inverse is also a function.

b. Determine the inverse of this function. Is the inverse also a function?

x	1	-2	-1	0	2	3	4	-3
f (x)	2	0	3	-1	1	-2	5	1

Answser: Swap the x and y variables to create the inverse. Since function f was not a one-to-one function (the y value of 1 was used twice), the inverse will NOT be a function (because the x value of 1 now gets mapped to two separate y values which is not possible for functions).

x	2	0	3	-1	1	-2	5	1
$f^{-1}(x)$	1	-2	-1	0	2	3	4	-3

Find the inverse of the function $f(x) = x - 4$

Answer:

$y = x - 4$

$x = y - 4$

$x + 4 = y$

$f^{-1}(x) = x + 4$

Remember:
Set = y.
Swap the variables.
Solve for y.

"Is it a function?" - Quick answer without the graph

Think of all the graphing that you've done so far. The simplest method is to solve for "y =", make a T-chart, pick some values for x, solve for the corresponding values of y, plot your points, and connect the dots, yadda, yadda, yadda. Not only is this useful for graphing, but this methodology gives yet another way of identifying functions: If you can solve for "y =", then it's a function. In other words, if you can enter it into your graphing calculator, then it's a function. (The calculator can only handle functions.) For example, 2y + 3x = 6 is a function, because you can solve for y:

$$2y + 3x = 6$$
$$2y = -3x + 6$$
$$y = (-3/2)x + 3$$

On the other hand, $y^2 + 3x = 6$ is not a function, because you can not solve for a unique y:

$$y^2 + 3x = 6$$
$$y^2 = -3x + 6$$
$$y = \pm\sqrt{-3x + 6}$$

I mean, yes, this is solved for "y =", but it's not unique. Do you take the positive square root, or the negative? Besides, where's the "±" key on your graphing calculator? So, in this case, the relation is not a function. (You can also check this by using our first definition from above. Think of "x = −1". Then we get $y^2 - 3 = 6$, so $y^2 = 9$, and then y can be either −3 or +3. That is, if we did an arrow chart, there would be two arrows coming from x = −1.)

Definition: One-one, Onto, Bijection

- A function f from A to B is called one to one (or one- one) if whenever f(a) = f(b) then a = b. Such functions are also called injections.
- A function f from A to B is called onto if for all b in B there is an a in A such that f(a) = b. Such functions are also called surjections.
- A function f from A to B is called a bijection if it is one to one and onto, i.e. bijections are functions that are injective and surjective.

Examples:

- If the graph of a function is known, how can you decide whether a function is one-to-one (injective) or onto (surjective) ?
- Which of the following functions are one-one, onto, or bijections ? The domain for all functions is R.
    1. f(x) = 2x + 5
    2. g(x) = arctan(x)
    3. g(x) = sin(x)
    4. $h(x) = 2x^3 + 5x^2 - 7x + 6$

Bijection, injection and surjection

In mathematics, injections, surjections and bijections are classes of functions distinguished by the manner in which arguments (input expressions from the domain) and images (output expressions from the codomain) are related or mapped to each other.

Q-8: Let the function $\mathbf{f}:\mathfrak{R}^{\#}\to\mathfrak{R}^{\#}$ be defined by $\mathbf{f}(\mathbf{x})=\mathbf{x}^2$. Find:

(1) $\mathbf{f}^{-1}(25)$, (2) $\mathbf{f}^{-1}(-9)$, (3) $\mathbf{f}^{-1}([-1,1])$, (4) $\mathbf{f}^{-1}((-\infty,0))$. (5) $\mathbf{f}^{-1}([4,25])$.

Solution:

(1) $f^{-1}(25)=\{5,-5\}$ Since $f(5)=25$ and $f(-5)=25$ and since the square of no other number is 25.

(2) $\mathbf{f}^{-1}(-9)=\phi$ Since there is no real number whose square is -9, i.e. the equation $x^2=-9$ has no real root.

(3) $\mathbf{f}^{-1}([-1,1])=[-1,1]$ Since $|\mathbf{x}|\le\mathbf{1}$ implies $|\mathbf{x}^2|\le\mathbf{1}$, i.e. if x belongs to $[-1,1]$ then $f(x)=x^2$ also belongs to $[-1,1]$.

(4) $\mathbf{f}^{-1}((-\infty,0])=\{0\}$ Since $\mathbf{0}^2=\mathbf{0}\varepsilon(-\infty,\mathbf{0}]$ and since no other number squared belongs to $(-\infty,\mathbf{0}]$.

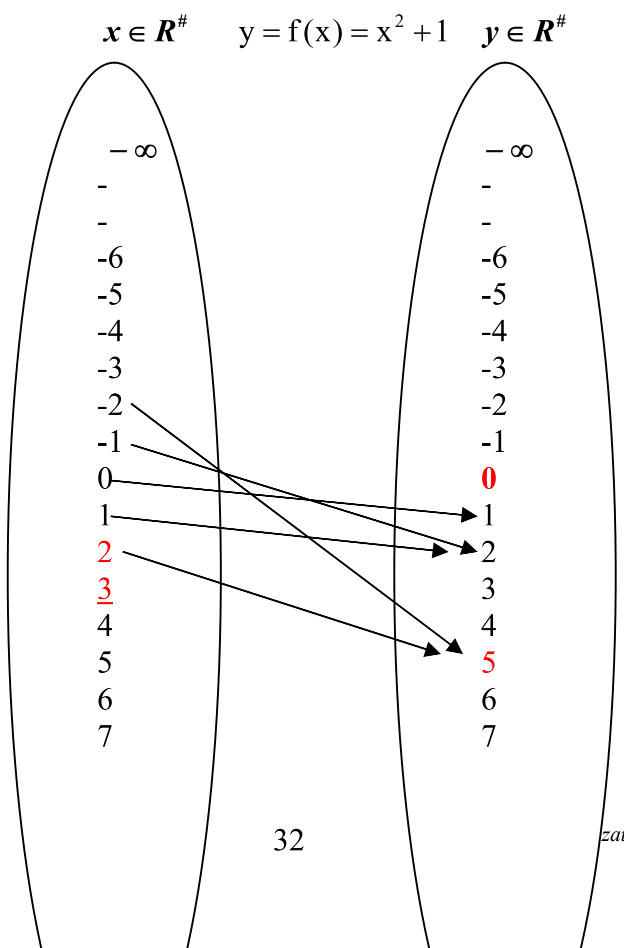
(5) $\mathbf{f}^{-1}([4,25])$ Consists of those numbers whose squares belong to $[4,25]$, i.e. those numbers $\mathbf{x}$ such that $\mathbf{4}\le\mathbf{x}^2\le\mathbf{25}$. Hence

$$\mathbf{f}^{-1}([4,25]) = \{\mathbf{x}|\ \ 2\le\mathbf{x}\le 5\ \text{ or }\ -5\le\mathbf{x}\le -2\}$$

Q-9: Let the function $\mathbf{f}:\mathfrak{R}^{\#}\to\mathfrak{R}^{\#}$ be defined by $\boldsymbol{y}=\boldsymbol{f}(\boldsymbol{x})=\boldsymbol{x}^2+\mathbf{1}$ Find:

(1) $\mathbf{f}^{-1}(5)$, (2) $\mathbf{f}^{-1}(0)$, (3) $\mathbf{f}^{-1}(10)$, (4) $\mathbf{f}^{-1}(-5)$. (5) $\boldsymbol{f}^{-1}([10,26])$. (6) $\mathbf{f}^{-1}([0,5])$.
(7) $\mathbf{f}^{-1}([-5,1])$. (8) $\mathbf{f}^{-1}([-5,5])$.

Solution:



8	8
9	9
10	10
11	11
12	12
13	
-	-
-	-
∞	∞

1) $f^{-1}(\underset{y}{\underset{\downarrow}{5}}) = \{x \in R^{\#};\ y = 5\}$

$= \{x \in R^{\#};\ x^2 + 1 = 5\}$

$= \{x \in R^{\#};\ x^2 = 4\}$

$= \{x \in R^{\#};\ x = \pm 2\}$

$= \{-2, 2\}$ Answer

2) $f^{-1}(\underset{y}{\underset{\downarrow}{0}}) = \{x \in R^{\#};\ y = 0\}$

$= \{x \in R^{\#};\ x^2 + 1 = 0\}$

$= \{x \in R^{\#};\ x^2 = -1\}$

$= \{x \in R^{\#};\ x = \pm\sqrt{-1}\}$

$= \{x \in R^{\#};\ x = \pm\sqrt{i^2}\}$

$= \{x \in R^{\#};\ x = \pm i\}$

$= \phi$ (Since $\pm\sqrt{-1}$ are not real numbers.) Answer

3) $f^{-1}(10) = \{x \in R^{\#};\ x^2 + 1 = 10\}$

$= \{x \in R^{\#};\ x^2 = 9\}$

$= \{x \in R^{\#};\ x = \pm 3\}$

$= \{-3, 3\}$ Answer

4) $f^{-1}(-5) = \{x \in R^{\#};\ x^2 + 1 = -5\}$

$= \{x \in R^{\#};\ x^2 = -6\}$

$= \{x \in R^{\#};\ x = \pm\sqrt{-6}\}$

$= \phi$ (Since $\pm\sqrt{-6}$ are not real numbers.) Answer

5) 4) $f^{-1}(\underset{y}{\underset{\downarrow}{[10, 26]}}) = \{x \in R^{\#};\ y \in [10, 26]\}$

$= \{x \in R^{\#};\ 10 \le y \le 26\}$

$= \{x \in R^{\#};\ 10 \le x^2 + 1 \le 26\}$

$= \{x \in R^{\#};\ 9 \le x^2 \le 25\}$

$= \{x \in R^{\#};\ \pm 3 \le x \le \pm 5\}$
$= \{x |\ 3 \le x \le 5,\ -5 \le x \le -3\}$

6) $f^{-1}([0,5]) = \{x \in R^{\#};\ f(x) \in [0,5]\}$
$= \{x \in R^{\#};\ 0 \le x^2 + 1 \le 5\}$
$= \{x \in R^{\#};\ -1 \le x^2 \le 4\}$
$= \{x \in R^{\#};\ \pm\sqrt{-1} \le x \le \pm 2\}$
$= \{x \in R^{\#};\ x \le \pm 2\}$

Since $\pm\sqrt{-1}$ are not real numbers.
$= \{x |\ -2 \le x \le 2\}$ Answer

7) $f^{-1}([-5,1]) = \{x \in R^{\#};\ f(x) \in [-5,1]\}$
$= \{x \in R^{\#};\ -5 \le x^2 + 1 \le 1\}$
$= \{x \in R^{\#};\ -6 \le x^2 \le 0\}$
$= \{x \in R^{\#};\ \pm\sqrt{-6} \le x \le \pm 0\}$
$= \{0\}$ Answer

Since $\pm\sqrt{-6}$ are not real numbers.

7) $f^{-1}([-5,5]) = \{x \in R^{\#};\ f(x) \in [-5,5]\}$
$= \{x \in R^{\#};\ -5 \le x^2 + 1 \le 5\}$
$= \{x \in R^{\#};\ -6 \le x^2 \le 4\}$
$= \{x \in R^{\#};\ \pm\sqrt{-6} \le x \le \pm 2\}$
$= \{x \in R^{\#};\ x \le \pm 2\}$

Since $\pm\sqrt{-6}$ are not real numbers.
$= \{x |\ -2 \le x \le 2\}$ Answer

Q-10: Let the function $f : \Re^{\#} \to \Re^{\#}$ be defined by y= $f(x) = \sin x$ Find:
(1) $f^{-1}(0)$, (2) $f^{-1}(1)$, (3) $f^{-1}(2)$. 4) $f^{-1}([-1,1])$.

Solution:

1) $f^{-1}(0) = \{x \in R^{\#};\ f(x) = 0\}$

$= \{x \in R^{\#};\ \sin x = 0\}$
$= \{x \in R^{\#};\ \sin x = \sin 0, \sin \pi, \sin 2\pi, -\sin 4\pi, -\sin 2\pi, ------\}$
$= \{x \in R^{\#};\ x = 0, \pi, 2\pi, -4\pi, -2\pi, ------\}$
$= \{-4\pi, -2\pi, -\pi,\ 0,\ \pi\ 2\pi, 4\pi, ------\}$

2) $f^{-1}(1) = \{x \in R^{\#};\ f(x) = 1\}$

$= \{x \in R^{\#};\ \sin x = 1\}$

$= \{x \in R^{\#};\ \sin x = \sin\frac{\pi}{2}, \sin\frac{5\pi}{2}, ------\}$

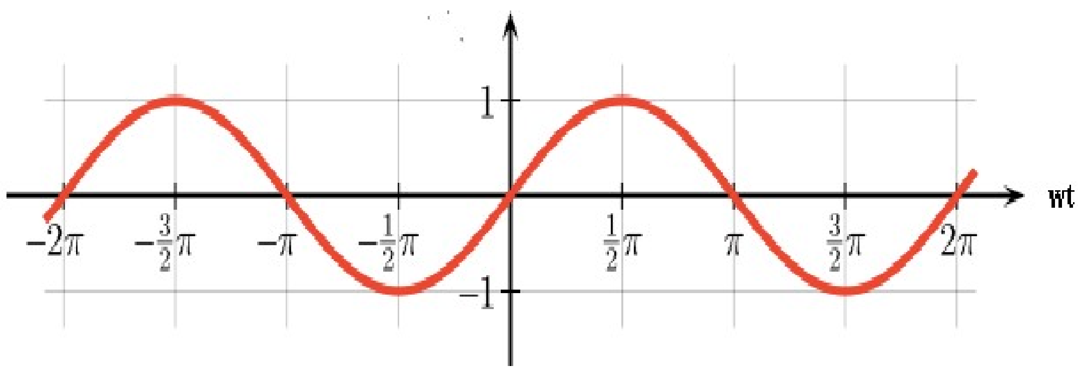
$= \{x \in R^{\#};\ x = \frac{\pi}{2}, \frac{5\pi}{2}, ------\}$

$= \{x \mid x = \frac{\pi}{2} + 2\pi n, \text{ Where } n \in z\}$ Answer

3) $f^{-1}(2) = \{x \in R^{\#};\ f(x) = 2\}$

$= \{x \in R^{\#};\ \sin x = 2\}$

$= \phi$



4) $f^{-1}([-1,1])$ = The set of all real numbers.

Q-11: Let the function $f: \Re^{\#} \to \Re^{\#}$ be defined by $f(x) = x^2 + x - 2$ Find:
(1) $f^{-1}(10)$, (2) $f^{-1}(4)$, (3) $f^{-1}(-5)$.

Solution:

1) $f^{-1}(10) = \{x \in R^{\#};\ x^2 + x - 2 = 10\}$

$= \{x \in R^{\#};\ x^2 + x - 12 = 0\}$

$= \{x \in R^{\#};\ x = \frac{-1 \pm \sqrt{1+48}}{2}\}$

$= \{x \in R^{\#};\ x = \frac{-1 \pm \sqrt{49}}{2}\}$

$= \{x \in R^{\#};\ x = \frac{-1 \pm 7}{2}\}$

$= \{x \in R^{\#};\ x = -4, 3\}$

$= \{-4, 3\}$ Answer

2) $f^{-1}(4) = \{x \in R^{\#};\ \ x^2 + x - 2 = 4\}$

$= \{x \in R^{\#};\ \ x^2 + x - 6 = 0\}$

$= \{x \in R^{\#};\ \ x = \dfrac{-1 \pm \sqrt{1+24}}{2}\}$

$= \{x \in R^{\#};\ \ x = \dfrac{-1 \pm \sqrt{25}}{2}\}$

$= \{x \in R^{\#};\ \ x = \dfrac{-1 \pm 5}{2}\}$

$= \{x \in R^{\#};\ \ x = -3, 2\}$

$= \{-3, 2\}$ Answer

3) $f^{-1}(-5) = \{x \in R^{\#};\ \ x^2 + x - 2 = -5\}$

$= \{x \in R^{\#};\ \ x^2 + x + 3 = 0\}$

$= \{x \in R^{\#};\ \ x = \dfrac{-1 \pm \sqrt{1-12}}{2}\}$

$= \{x \in R^{\#};\ \ x = \dfrac{-1 \pm \sqrt{-11}}{2}\}$

$= \phi$

Since $\dfrac{-1 \pm \sqrt{-11}}{2}$ are not real numbers.

Q-12: Let $A = R^{\#} - \{-\frac{1}{2}\}$ and $B = R^{\#} - \{\frac{1}{2}\}$, Let $f : A \to B$ be defined by $f(x) = \dfrac{(x-3)}{(2x+1)}$, Then f is one-one and onto. Find a formula that defines f^{-1}

Solution:

1) Domain = Set of all real numbers except $x = -\frac{1}{2}$, i.e. $R^{\#} - \{-\frac{1}{2}\}$

2) Let $y = f(x) = \dfrac{(x-3)}{(2x+1)}$

$\Rightarrow 2xy + y = x - 3$

$\Rightarrow x = \dfrac{-3-y}{2y-1}$

This is undefined for $y = \frac{1}{2}$

Hence Range = $R^{\#} - \{\frac{1}{2}\}$

$\mathbf{R} = \{(\mathbf{1},\mathbf{a}), \quad (\mathbf{1},\mathbf{b}) \quad (\mathbf{3},\mathbf{a})\}$ is a relation from A to B.
The inverse relation of R is $\mathbf{R}^{-1} = \{(\mathbf{a},\mathbf{1}), \quad (\mathbf{b},\mathbf{1}), \quad (\mathbf{a},\mathbf{3})\}$

Example 2: Let $\mathbf{W} = \{\mathbf{a},\mathbf{b},\mathbf{c}\}$. Then
$\mathbf{R} = \{(\mathbf{a},\mathbf{b}), \quad (\mathbf{a},\mathbf{c}), \quad (\mathbf{c},\mathbf{c}) \quad (\mathbf{c},\mathbf{b})\}$ is a relation in W. The inverse relation of R is
$\mathbf{R}^{-1} = \{(\mathbf{b},\mathbf{a}), \quad (\mathbf{c},\mathbf{a}), \quad (\mathbf{c},\mathbf{c}) \quad (\mathbf{b},\mathbf{c})\}$

1. **Reflexive:** Each element is related to itself.

2. **Symmetric:** If any one element is related to any other element, then the second element is related to the first.

3. **Transitive:** If any one element is related to a second and that second element is related to a third, then the first element is related to the third.

Reflexive Relations:

Let $\mathbf{R} = (\mathbf{A},\mathbf{A},\mathbf{P}(\mathbf{x},\mathbf{y}))$ be a relation in a set A, i.e. Let R be a subset of $\mathbf{A} \times \mathbf{A}$. Then R is called a reflexive relation if, for every $\mathbf{a} \in \mathbf{A}, \quad (\mathbf{a},\mathbf{a}) \in \mathbf{R}$

In other words, R is reflexive if every element in A is related to itself.

Example 1: Let $\mathbf{V} = \{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4}\}$ and $\mathbf{R} = \{(\mathbf{1},\mathbf{1}), \quad (\mathbf{2},\mathbf{4}), \quad (\mathbf{3},\mathbf{3}), \quad (\mathbf{4},\mathbf{1}), \quad (\mathbf{4},\mathbf{4})\}$
Then R is not a reflexive relation since (2, 2) does not belong to R. Notice that all ordered pairs (a, a) must belong to R in order for R to be reflexive.

Example 2: Let R be the relation in the real numbers defined by the open sentence "x is less than y", i.e. "x<y". Then R is not reflexive since $a < a$ for any real number a.

Example 3: Let W = {1, 2, 3, 4} and R = {(1, 1), (1, 3), (2, 2), (3, 1), (4, 4)}. Is R reflexive?
Solution: R is not reflexive since $\mathbf{3} \in \mathbf{W}$ and $(\mathbf{3},\mathbf{3}) \notin \mathbf{R}$.

Example 4: Let. E = {1, 2, 3}. Consider the following relations in E:

R_1 = {(1, 2), (3, 2), (2, 2), (2, 3)} $\qquad$ R_4 = {(1, 2)}
R_2 = {(1, 2), (2, 3), (1, 3) $\qquad$ R_5 = E × E
R_3 = {(1, 1), (2, 2), (2, 3), (3, 2), (3, 3)}

State whether or not each of these relations is reflexive.
Solution: If a relation in E is reflexive, then (1, 1), (2, 2) and (3, 3) must belong to the relation. Therefore only R_3 and R_5 are reflexive.

Symmetric Relations:

Let R be a subset of $\mathbf{A} \times \mathbf{A}$, i.e. let R be a relation in A. Then R is called a symmetric relation if $(\mathbf{a},\mathbf{b}) \in \mathbf{R}$ implies $(\mathbf{b},\mathbf{a}) \in \mathbf{R}$

That is, if a is related to b then b is also related to a.

Example 1: When is a relation R in a set A not symmetric?

Solution: R is not symmetric if there are elements $\mathbf{a} \in \mathbf{A}, \quad \mathbf{b} \in \mathbf{A}$ such that $(\mathbf{a},\mathbf{b}) \in \mathbf{R}, \quad (\mathbf{b},\mathbf{a}) \notin \mathbf{R}$

Note $a \neq b$, otherwise $(a,b) \in R$ implies $(\mathbf{b},\mathbf{a}) \in \mathbf{R}.$

Example 2: Let V = {1, 2, 3, 4} and R = {(1,2), (3,4), (2,1), (3,3)}. Is R symmetric?

Solution: R is not symmetric, since $\mathbf{3} \in \mathbf{V}, \quad \mathbf{4} \in \mathbf{V}, \quad (\mathbf{3},\mathbf{4}) \in \mathbf{R}$ and $(\mathbf{4},\mathbf{3}) \notin \mathbf{R}.$

Example 3: Let E = {1, 2, 3}. Consider the following relations in E:

$R_1 = \{(1,1), (2,1), (2,2), (3,2), (2,3)\}$ $\quad R_4 = \{(1,1), (3,2), (2,3)\}$

$R_2 = \{(1, 1)\}$ $\quad R_5 = E \times E$

$R_3 = \{(1, 2)\}$

State whether or not each of these relations is symmetric.

Solution:

(1) R_1 is not symmetric since (2, 1) $\in R_1$ but (1, 2) $\notin R_1$.

(4) R_4 is symmetric.

(2) R_2 is symmetric.

(5) R_5 is symmetric.

(3) R_3 is not symmetric since (1, 2) $\in \mathbf{R}_3$ but (2, 1) $\notin \mathbf{R}_3$.

Anti-Symmetric Relations:

A relation R in a set A, i.e. a subset of $\mathbf{A} \times \mathbf{A}$, is called an anti-symmetric relation

i. if $(\mathbf{a},\mathbf{b}) \in \mathbf{R}$ and $(\mathbf{b},\mathbf{a}) \in \mathbf{R}$ implies a = b.

ii. In other words, if $\mathbf{a} \neq \mathbf{b}$ then possibly a is related to b or possibly b is related to a, but never both.

Example 1: Let N be the natural numbers and let R be the relation in N defined by "x divides. y". Then R is anti-symmetric since a divides b and b divides a implies a = b

Example 2: Let $\mathbf{W} = \{1,2,3,4\}$, let $\mathbf{R} = \{(1,3), \quad (4,2), \quad (4,4), \quad (2,4)\}$

Then R is not an anti-symmetric relation in W since $(4,2) \in \mathbf{R}$ and $(2,4) \in \mathbf{R}$

Example 3: Where is a relation R in a set A not anti-symmetric?

Solution:

R is not anti-symmetric if there exists elements $\mathbf{a} \in \mathbf{A}, \mathbf{b} \in \mathbf{A}, \mathbf{a} \neq \mathbf{b}$ such that $(\mathbf{a},\mathbf{b}) \in \mathbf{R}$ and $(\mathbf{b},\mathbf{a}) \in \mathbf{R}.$

Example 4: Let $\mathbf{W} = \{1,2,3,4\}$ and $\mathbf{R} = \{(1,2), \quad (3,4), \quad (2,2), \quad (3,3), \quad (2,1)\}$. Is R anti-symmetric?
Solution: R is not anti-symmetric, science $\mathbf{1 \in W, 2 \in W, 1 \neq 2, (1,2) \in R}$ and $(\mathbf{2,1}) \in \mathbf{R}.$

Example 5: Let E = {1, 2, 3}. Consider the following relations in E:

R_1 = {(1,1), (2,1), (2,2), (3,2), (2,3)} R_4 = {(1,1), (2,3), (3,2)}
R_2 = {(1, 1)} R_5 = E × E
R_3 = {(1, 2)}

State whether or not each of these relations is anti-symmetric.
Solution:
(1) R_1 is not anti-symmetric since (3, 2) $\in \mathbf{R_1}$ and (2, 3) $\in \mathbf{R_1}$.
(2) R_2 is anti-symmetric.
(3) R_3 is anti-symmetric.
(4) R_4 is not symmetric since (2, 3) $\in \mathbf{R_4}$ and (3, 2) $\in \mathbf{R_4}$.
(5) R_5 is not anti-symmetric for the same reasons as for R_4.

TRANSITIVE RELATIONS

A relation R in a set A is called a transitive relation if $(\mathbf{a,b}) \in \mathbf{R}$ and $(\mathbf{b,c}) \in \mathbf{R}$ implies $(\mathbf{a,c}) \in \mathbf{R}$.

In other words, if a is related to b and b is related to c, then a is related to c.

Example 1: When is a relation in a set A not transitive?
Solution: R in not transitive if there exist elements a, b and c belonging to A, not necessarily distinct, such that $(\mathbf{a,b}) \in \mathbf{R}, \quad (\mathbf{b,c}) \in \mathbf{R}$ but $(\mathbf{a,c}) \notin \mathbf{R}$

Example 2: Let W = {1, 2, 3, 4} and R = {(1,2), (4,3), (2,2), (2,1), (3,1)}. Is R transitive?
Solution: R is not transitive since $(\mathbf{4,3}) \in \mathbf{R}, \quad (\mathbf{3,1}) \in \mathbf{R}$ but $(\mathbf{4,1}) \notin \mathbf{R}.$

Example 3: Let W = {1, 2, 3, 4} and R = {(2,2), (2,3), (1,4), (3,2)}. Is R transitive?
Solution: R is not transitive since $(\mathbf{3,2}) \in \mathbf{R}, \quad (\mathbf{2,3}) \in \mathbf{R}$ but $(\mathbf{3,3}) \notin \mathbf{R}.$

Example 4: Let E = $\{\mathbf{1,2,3}\}$. Consider the following relations in E:
R_1 = {(1, 2), (2, 2)}
R_4 = {(1, 1)}
R_2 = {(1,2), (2,3), (1,3), (2,1), (1,1)}
R_5 = E × E
R_3 = {(1, 2)}
State whether or not each of these relations is transitive.
Solution:

Each of the relations is transitive except R_2.
R_2 is not transitive, since

$$(2,1)\in \mathbf{R}_2,(1,2)\in \mathbf{R}_2 \quad \text{But } (2,2)\notin \mathbf{R}_2$$

Example: Let. E = {a, b, c}. Consider the following relations in E:
R = {(a, a), (b, a), (b, b), (c, b), (b,c)}

State whether or not each of these relations is reflexive, Symmetric, Anti-Symmetric, and Transitive.

Answer:

1. R is not reflexive since $(\mathbf{c},\mathbf{c}) \notin \mathbf{R}$
2. R is not symmetric since $(\mathbf{b},\mathbf{a}) \in \mathbf{R}$ but $(\mathbf{a},\mathbf{b}) \notin \mathbf{R}$
3. R is not anti-symmetric since $(\mathbf{c},\mathbf{b}) \in \mathbf{R}$, $(\mathbf{b},\mathbf{c}) \in \mathbf{R}$ but $b \neq c$
4. R is not transitive since $(\mathbf{c},\mathbf{b}) \in \mathbf{R}$, $(\mathbf{b},\mathbf{c}) \in \mathbf{R}$ but $(\mathbf{c},\mathbf{c}) \notin \mathbf{R}$

Example: Let. E = {1, 2, 3}. Consider the following relations in E:
R = {(1, 1), (1, 2)}

State whether or not each of these relations is reflexive, Symmetric, Anti-Symmetric, and Transitive.

Answer:

5. R is not reflexive since $(\mathbf{2},\mathbf{2}) \notin \mathbf{R}$, $(\mathbf{3},\mathbf{3}) \notin \mathbf{R}$
6. R is not symmetric since $(\mathbf{1},\mathbf{2}) \in \mathbf{R}$ but $(\mathbf{2},\mathbf{1}) \notin \mathbf{R}$
7. R is anti-symmetric since $\mathbf{1} \neq \mathbf{2}$, $(\mathbf{1},\mathbf{2}) \in \mathbf{R}$,
8. R is transitive since $(\mathbf{1},\mathbf{1}) \in \mathbf{R}$, $(\mathbf{1},\mathbf{2}) \in \mathbf{R}$ implies $(\mathbf{1},\mathbf{2}) \in \mathbf{R}$

Example: Let. E = {1, 2, 3, 4}. Consider the following relations in E:
R = {(1, 1), (2, 3), (4, 1)}

State whether or not each of these relations is reflexive, Symmetric, Anti-Symmetric, and Transitive.

Answer:

9. R is not reflexive since $(\mathbf{2},\mathbf{2}) \notin \mathbf{R}$, $(\mathbf{3},\mathbf{3}) \notin \mathbf{R}$, $(\mathbf{4},\mathbf{4}) \notin \mathbf{R}$
10. R is not symmetric since $(\mathbf{2},\mathbf{3}) \in \mathbf{R}$ but $(\mathbf{3},\mathbf{2}) \notin \mathbf{R}$, $(\mathbf{4},\mathbf{1}) \in \mathbf{R}$ but $(\mathbf{1},\mathbf{4}) \notin \mathbf{R}$
11. R is anti-symmetric since $\mathbf{2} \neq \mathbf{3}$ $(\mathbf{2},\mathbf{3}) \in \mathbf{R}$, $\mathbf{4} \neq \mathbf{1}$ $(\mathbf{4},\mathbf{1}) \in \mathbf{R}$
12. R is transitive since $(\mathbf{1},\mathbf{1}) \in \mathbf{R}$, $(\mathbf{4},\mathbf{1}) \in \mathbf{R}$, $(\mathbf{2},\mathbf{3}) \in \mathbf{R}$

Equivalence Relations:

A relation R in a set A is an equivalence relation if

i) R is reflexive, that is, for every $\mathbf{a} \in \mathbf{R}$, $(\mathbf{a},\mathbf{a}) \in \mathbf{R}$,

Demoivre's Theorem

State and Prove Demoivre's theorem

Statement: whatever be the value of n, positive or negative, integral or fractional,

$\mathbf{\cos n\theta + i\sin n\theta}$ is the value or one of the values of $\mathbf{(\cos\theta + i\sin\theta)^n}$

Proof:

Case 1: when n is a positive integer,

By actual multiplication $(x_1+iy_1)(x_2+iy_2) = r_1(\cos\theta_1 + i\sin\theta_1)r_2(\cos\theta_2 + i\sin\theta_2)$

$\therefore r_1(\cos\theta_1 + i\sin\theta_1)r_2(\cos\theta_2 + i\sin\theta_2)$

$\therefore r_1r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$

$= r_1r_2\{\cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 + i^2\sin\theta_1\sin\theta_2\}$

$= r_1r_2\{\cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2\}$ $\qquad [i^2 = -1]$

$= r_1r_2\{\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2\}$

$[\cos A\cos B - \sin A\sin B = \cos(A+B)\ \&\ \sin A\cos B + \cos A\sin B = \sin(A+B)]$

$= r_1r_2\{\cos(\theta_1+\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\}$

$= r_1r_2\{\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\}$

$\therefore r_1r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) = r_1r_2\{\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\}$

$\therefore (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) = \{\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\}$ ----------------(i)

Now, $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3)$

$= \{\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\}(\cos\theta_3 + i\sin\theta_3)$

$= \cos(\theta_1+\theta_2)\cos\theta_3 + i\sin(\theta_1+\theta_2)\cos\theta_3 + i\cos(\theta_1+\theta_2)\sin\theta_3 + i^2\sin(\theta_1+\theta_2)\sin\theta_3$

$= \cos(\theta_1+\theta_2)\cos\theta_3 + i\sin(\theta_1+\theta_2)\cos\theta_3 + i\cos(\theta_1+\theta_2)\sin\theta_3 - \sin(\theta_1+\theta_2)\sin\theta_3$

$= \cos(\theta_1+\theta_2)\cos\theta_3 - \sin(\theta_1+\theta_2)\sin\theta_3 + i\{\sin(\theta_1+\theta_2)\cos\theta_3 + \cos(\theta_1+\theta_2)\sin\theta_3\}$

$= \cos(\theta_1+\theta_2+\theta_3) + i\{\sin(\theta_1+\theta_2+\theta_3\}$

So, $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3)\ldots\ldots\ldots\ldots(\cos\theta_n + i\sin\theta_n)$

$= \cos(\theta_1+\theta_2+\theta_3+\ldots\ldots\ldots+\theta_n) + i\{\sin(\theta_1+\theta_2+\theta_3+\theta_4\ldots\ldots\ldots+\theta_n\}$

If $\theta_1 = \theta_2 = \theta_3 = \ldots\ldots\ldots\ldots\ldots = \theta_n = \theta$

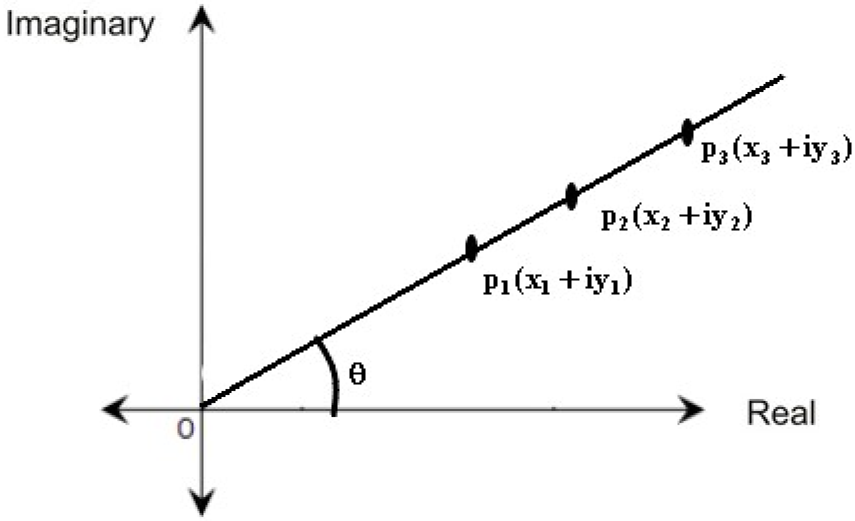


Figure 3

Then $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3)......................(\cos\theta_n + i\sin\theta_n)$

$= \cos(\theta_1 + \theta_2 + \theta_3 + ... + \theta_n) + i\{\sin(\theta_1 + \theta_2 + \theta_3 + \theta_3 .... + \theta_n\}$

$\Rightarrow (\cos\theta + i\sin\theta)(\cos\theta + i\sin\theta)(\cos\theta + i\sin\theta)......................(\cos\theta + i\sin\theta)$

$= \cos(\theta + \theta + \theta + ... + \theta) + i\{\sin(\theta + \theta + \theta + \theta .... + \theta\}$

$\Rightarrow (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

Case 2: when n is a negative integer

Let us suppose, $n = -m$, where m is a positive integer.

$\Rightarrow (\cos\theta + i\sin\theta)^n = (\cos\theta + i\sin\theta)^{-m}$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{1}{(\cos\theta + i\sin\theta)^m}$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{1}{\cos m\theta + i\sin m\theta}$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{(\cos m\theta - i\sin m\theta)}{(\cos m\theta + i\sin m\theta)(\cos m\theta - i\sin m\theta)}$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{(\cos m\theta - i\sin m\theta)}{\cos^2 m\theta - i^2\sin^2 m\theta}$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{(\cos m\theta - i\sin m\theta)}{\cos^2 m\theta + \sin^2 m\theta} \qquad [i^2 = -1]$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \frac{(\cos m\theta - i\sin m\theta)}{1} \qquad [\cos^2\theta + \sin^2\theta = 1]$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = (\cos m\theta - i\sin m\theta)$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \cos(-m\theta) + i\sin(-m\theta) \qquad [\cos(-\theta) = \cos\theta;\ \sin(-\theta) = -\sin\theta]$$

$$\Rightarrow (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \qquad [\because n = -m]$$

Case 3: when n is a fraction, positive or negative

Let us suppose, $\mathbf{n = \frac{p}{q}}$, where q is a positive integer and p is any integer, positive or negative.

From case 1:

$$(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q})^q = \cos q\frac{\theta}{q} + i\sin q\frac{\theta}{q} \qquad [(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta]$$

$$(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q})^q = \cos\theta + i\sin\theta$$

Taking the q-th roots on both sides,

$$(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q})^q = \cos\theta + i\sin\theta$$

$$\{(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q})^q\}^{\frac{1}{q}} = (\cos\theta + i\sin\theta)^{\frac{1}{q}}$$

$$(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q}) = (\cos\theta + i\sin\theta)^{\frac{1}{q}}$$

So, $\cos\frac{\theta}{q} + i\sin\frac{\theta}{q}$ is one of the values of $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$

Raising to the p-th power,

$$(\cos\frac{\theta}{q} + i\sin\frac{\theta}{q}) = (\cos\theta + i\sin\theta)^{\frac{1}{q}}$$

$$\{(\cos\frac{\theta}{q}+i\sin\frac{\theta}{q})\}^p=\{(\cos\theta+i\sin\theta)^{\frac{1}{q}}\}^p$$

$$\{(\cos\frac{\theta}{q}+i\sin\frac{\theta}{q})\}^p=\{(\cos\theta+i\sin\theta)^{\frac{p}{q}}\}$$

$$\{(\cos\frac{\theta}{q}+i\sin\frac{\theta}{q})\}^p=\{(\cos\theta+i\sin\theta)^{\frac{p}{q}}\}$$

$$\Rightarrow \cos p\frac{\theta}{q}+i\sin p\frac{\theta}{q}=\{(\cos\theta+i\sin\theta)^{\frac{p}{q}}\}$$

$$\Rightarrow \cos\frac{p}{q}\theta+i\sin\frac{p}{q}\theta=\{(\cos\theta+i\sin\theta)^{\frac{p}{q}}\}$$

$$\Rightarrow (\cos\theta+i\sin\theta)^{\frac{p}{q}}=\cos\frac{p}{q}\theta+i\sin\frac{p}{q}\theta$$

$$\Rightarrow (\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta \qquad [n=\frac{p}{q}]$$

Q-01: If $x_r=\cos\frac{\pi}{3^r}+i\sin\frac{\pi}{3^r}$, prove that $x_1x_2x_3\ldots\ldots\ldots\ldots\inf=i$

Answer: Given,

$$x_r=\cos\frac{\pi}{3^r}+i\sin\frac{\pi}{3^r}$$

Putting r = 1, 2, 3, 4, …………..

$$x_1=\cos\frac{\pi}{3^1}+i\sin\frac{\pi}{3^1}$$

$$x_2=\cos\frac{\pi}{3^2}+i\sin\frac{\pi}{3^2}$$

$$x_3=\cos\frac{\pi}{3^3}+i\sin\frac{\pi}{3^3}$$

$$x_4=\cos\frac{\pi}{3^4}+i\sin\frac{\pi}{3^4}$$

…………………………………….

……………………………………

………………………………………..

$\therefore x_1x_2x_3$ ………………

$(\cos\frac{\pi}{3^1}+i\sin\frac{\pi}{3^1})\ (\cos\frac{\pi}{3^2}+i\sin\frac{\pi}{3^2})\ (\cos\frac{\pi}{3^3}+i\sin\frac{\pi}{3^3})\ (\cos\frac{\pi}{3^4}+i\sin\frac{\pi}{3^4})$………….

$=\cos(\frac{\pi}{3^1}+\frac{\pi}{3^2}+\frac{\pi}{3^3}+\frac{\pi}{3^4}+\ldots\ldots\ldots\ldots)+i\sin(\frac{\pi}{3^1}+\frac{\pi}{3^2}+\frac{\pi}{3^3}+\frac{\pi}{3^4}+\ldots\ldots\ldots\ldots)$

$=\cos\{\frac{\pi}{3^1}(1+\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\ldots\ldots)\}+i\sin\{\frac{\pi}{3^1}(1+\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\ldots\ldots)\}$

$[\because (1-x)^{-1}=1+x+x^2+x^3+x^4+\ldots\ldots\ldots\ldots]$

$=\cos\{\frac{\pi}{3^1}(1-\frac{1}{3})^{-1}\}+i\sin\{\frac{\pi}{3^1}(1-\frac{1}{3})^{-1}\}$

$=\cos\{\frac{\pi}{3^1}(\frac{3-1}{3})^{-1}\}+i\sin\{\frac{\pi}{3^1}(\frac{3-1}{3})^{-1}\}$

$=\cos\{\frac{\pi}{3^1}(\frac{2}{3})^{-1}\}+i\sin\{\frac{\pi}{3^1}(\frac{2}{3})^{-1}\}$

$=\cos\{\frac{\pi}{3^1}(\frac{1}{\frac{2}{3}})\}+i\sin\{\frac{\pi}{3^1}(\frac{1}{\frac{2}{3}})\}$

$=\cos\{\frac{\pi}{3^1}(\frac{3}{2})\}+i\sin\{\frac{\pi}{3^1}(\frac{3}{2})\}$

$=\cos\{\frac{\pi}{2}\}+i\sin\{\frac{\pi}{2}\}$

$=0+i.1$

$=i$

Q-02: If $(1+i\frac{x}{a})(1+i\frac{x}{b})(1+i\frac{x}{c})\ldots\ldots\ldots=A+iB$, Then prove that

(i) $(1+\frac{x^2}{a^2})(1+\frac{x^2}{b^2})(1+\frac{x^2}{c^2})\ldots\ldots\ldots\ldots\ldots=A^2+B^2$

(ii) $\tan^{-1}\frac{x}{a}+\tan^{-1}\frac{x}{b}+\tan^{-1}\frac{x}{c}+\ldots\ldots\ldots\ldots=\tan^{-1}\frac{B}{A}$

Answer:

Given, $(1+i\frac{x}{a})(1+i\frac{x}{b})(1+i\frac{x}{c})\ldots\ldots\ldots = A+iB$ ------------------------------------(i)

Let, $1 = r\cos\alpha,\ \frac{x}{a} = r\sin\alpha$ Again, Let, $1 = r\cos\beta,\ \frac{x}{b} = r\sin\beta$ Again, Let,

$1 = r\cos\gamma,\ \frac{x}{c} = r\sin\gamma$

So, $\frac{\frac{x}{a}}{1} = \frac{r\sin\alpha}{r\cos\alpha}$

$\therefore \frac{x}{a} = \tan\alpha$

$\therefore \tan\alpha = \frac{x}{a}$

$\therefore \alpha = \tan^{-1}\frac{x}{a}$

So, $\frac{\frac{x}{b}}{1} = \frac{r\sin\beta}{r\cos\beta}$

$\therefore \frac{x}{b} = \tan\beta$

$\therefore \tan\beta = \frac{x}{b}$

$\therefore \beta = \tan^{-1}\frac{x}{b}$

So, $\frac{\frac{x}{c}}{1} = \frac{r\sin\gamma}{r\cos\gamma}$

$\therefore \frac{x}{c} = \tan\gamma$ etc. ,

$\therefore \tan\gamma = \frac{x}{c}$

$\therefore \gamma = \tan^{-1}\frac{x}{c}$

From (i), we get

Given, $(1+i\frac{x}{a})(1+i\frac{x}{b})(1+i\frac{x}{c})\ldots\ldots\ldots = A+iB$

$\Rightarrow (1+i\tan\alpha)(1+i\tan\beta)(1+i\tan\gamma)\ldots\ldots\ldots = A+iB$

$\Rightarrow (1+i\frac{\sin\alpha}{\cos\alpha})(1+i\frac{\sin\beta}{\cos\beta})(1+i\frac{\sin\gamma}{\cos\gamma})\ldots\ldots\ldots = A+iB$

$\Rightarrow (\frac{\cos\alpha+i\sin\alpha}{\cos\alpha})(\frac{\cos\beta+i\sin\beta}{\cos\beta})(\frac{\cos\gamma+i\sin\gamma}{\cos\gamma})\ldots\ldots\ldots = A+iB$

$\Rightarrow (\frac{1}{\cos\alpha})(\frac{1}{\cos\beta})(\frac{1}{\cos\gamma})\ldots\ldots\ldots.(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)(\cos\gamma+i\sin\gamma)\ldots\ldots\ldots = A+iB$

$\Rightarrow (\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)(\cos\gamma+i\sin\gamma)\ldots\ldots\ldots = A+iB$

$\Rightarrow (\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots[\cos(\alpha+\beta+\gamma+\ldots\ldots\ldots)+i\sin(\alpha+\beta+\gamma+\ldots\ldots\ldots)] = A+iB$

Equating the real and imaginary part on both sides, we get,

$(\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots[\cos(\alpha+\beta+\gamma+\ldots\ldots\ldots)] = A$ --------------------(ii)

$(\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots[\sin(\alpha+\beta+\gamma+\ldots\ldots\ldots)] = B$ --------------------(iii)

Squaring (ii) and (iii), we get,

$$(\sec^2\alpha)(\sec^2\beta)(\sec^2\gamma)\ldots\ldots\ldots[\cos^2(\alpha+\beta+\gamma+\ldots\ldots\ldots)]=A^2 \text{--------------------(iv)}$$

$$(\sec^2\alpha)(\sec^2\beta)(\sec^2\gamma)\ldots\ldots\ldots[\sin^2(\alpha+\beta+\gamma+\ldots\ldots\ldots)]=B^2 \text{--------------------(v)}$$

Adding (iv) & (v)

$$(\sec^2\alpha\sec^2\beta\sec^2\gamma\ldots.)\cos^2(\alpha+\beta+\gamma+\ldots.)+(\sec^2\alpha\sec^2\beta\sec^2\gamma\ldots)\sin^2(\alpha+\beta+\gamma+\ldots)=A^2+B^2$$

$$\Rightarrow(\sec^2\alpha\sec^2\beta\sec^2\gamma\ldots\ldots\ldots..)\{\cos^2(\alpha+\beta+\gamma+\ldots\ldots)+\sin^2(\alpha+\beta+\gamma+\ldots.)\}=A^2+B^2$$

$$\Rightarrow(\sec^2\alpha\sec^2\beta\sec^2\gamma\ldots\ldots\ldots..)1=A^2+B^2$$

$$\Rightarrow\sec^2\alpha\sec^2\beta\sec^2\gamma\ldots\ldots\ldots..=A^2+B^2$$

$$\Rightarrow(1+\tan^2\alpha)(1+\tan^2\beta)(1+\tan^2\gamma)\ldots\ldots\ldots..=A^2+B^2$$

$$\therefore(1+\frac{x^2}{a^2})(1+\frac{x^2}{b^2})(1+\frac{x^2}{c^2})\ldots\ldots\ldots..=A^2+B^2 \text{ proved (i)}$$

Now, (iii) $\div$ (ii)

$$\frac{(\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots[\sin(\alpha+\beta+\gamma+\ldots\ldots\ldots)]}{(\sec\alpha)(\sec\beta)(\sec\gamma)\ldots\ldots\ldots[\cos(\alpha+\beta+\gamma+\ldots\ldots\ldots)]}=\frac{B}{A}$$

$$\Rightarrow\frac{[\sin(\alpha+\beta+\gamma+\ldots\ldots\ldots)]}{[\cos(\alpha+\beta+\gamma+\ldots\ldots\ldots)]}=\frac{B}{A}$$

$$\Rightarrow\tan(\alpha+\beta+\gamma+\ldots\ldots\ldots)=\frac{B}{A}$$

$$\Rightarrow\alpha+\beta+\gamma+\ldots\ldots\ldots=\tan^{-1}\frac{B}{A}$$

$$\therefore(\tan^{-1}\frac{x}{a}+\tan^{-1}\frac{x}{b}+\tan^{-1}\frac{x}{c}+\ldots\ldots\ldots..)=\tan^{-1}\frac{B}{A} \text{ proved (ii)}$$

Q-03: Using Demoivre's theorem, solve the equation $x^6+x^5+x^4+x^3+x^2+x^1+1=0$

Answer: We have, $x^6+x^5+x^4+x^3+x^2+x^1+1=0$

Multiplying the given equation by (x-1), we get,

$$(x-1)(x^6+x^5+x^4+x^3+x^2+x^1+1)=0$$

$$\Rightarrow x^7-1=0$$

$$\Rightarrow x^7=1$$

cos(4.pi+0)

=cos(8.90 +0)

=cos0

=1

$$\Rightarrow x^7 = 1$$

$$\Rightarrow x = (1)^{1/7}$$

$$\Rightarrow x = (\cos 0 + i\sin 0)^{1/7}$$

$$\Rightarrow x = \{\cos(2n\pi + 0) + i\sin(2n\pi + 0)\}^{1/7}$$

$$\Rightarrow x = \cos\frac{2n\pi}{7} + i\sin\frac{2n\pi}{7} \qquad [(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta]$$

Putting n =0, 1, 2, 3, 4, 5 and 6, we get roots of equation as

$$\Rightarrow x = \cos 0 + i\sin 0$$

$$\Rightarrow x = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$$

$$\Rightarrow x = \cos\frac{4\pi}{7} + i\sin\frac{4\pi}{7}$$

$$\Rightarrow x = \cos\frac{6\pi}{7} + i\sin\frac{6\pi}{7}$$

$$\Rightarrow x = \cos\frac{8\pi}{7} + i\sin\frac{8\pi}{7}$$

$$\Rightarrow x = \cos\frac{10\pi}{7} + i\sin\frac{10\pi}{7}$$

$$\Rightarrow x = \cos\frac{12\pi}{7} + i\sin\frac{12\pi}{7}$$

Q-04: Using Demoivre's theorem, find the values of